



A comprehensive survey on pose estimation and tracking in sports: methodologies, datasets, challenges, and future directions

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Abstract

Pose estimation and tracking in sports has gained significant attention due to its potential to revolutionize performance analysis, injury prevention, and strategic decision-making. This survey presents a comprehensive overview of the methodologies, datasets, challenges, and future directions in this rapidly evolving field. We explore traditional approaches, including geometric and statistical models, and highlight the transformative impact of deep learning techniques, such as convolutional neural networks, transformers, and hybrid architectures, which have enabled highly accurate and robust pose estimation. The paper also discusses dataset creation and ground-truthing techniques tailored to sports contexts, emphasizing the importance of multimodal data, scalability, and representativeness. Applications across diverse sports, from individual to team-based activities, demonstrate the versatility of pose estimation systems in both real-time and offline settings. However, challenges such as occlusions, dynamic backgrounds, and computational efficiency persist, necessitating further innovation. We identify future research directions, including the integration of multimodal data, edge computing, and ethical considerations, to enhance accuracy, interpretability, and generalizability. This survey aims to provide a foundational reference for researchers and practitioners, fostering advancements in pose estimation and tracking technologies that meet the unique demands of sports analytics.

Keywords Pose estimation · Pose tracking · Sport science

1 Introduction

Pose estimation and tracking have become essential tools in modern sports (Bartlett 2014; Petrigna and Musumeci 2023) due to their transformative potential in performance analysis (Aderinola et al 2023; Li et al 2023; Xi et al 2024; Panconi et al 2024), injury prevention (Demircan et al 2009; Moustridi et al 2023; Johnson et al 2019), rehabilitation, strategic decision-making, and enhanced audience experience. By capturing detailed motion data,

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these systems allow coaches and athletes to identify inefficiencies in technique, refine training methods (Frangoudes et al 2022; Karg et al 2014), and optimize athletic performance. Biomechanics-based evaluation helps perfect a sprinter's stride and improve a gymnast's posture. It also offers valuable insights into movement patterns (Liu and Yuan 2018; Fang et al 2022; Raza et al 2023; Gozlan et al 2024), posture (Elforaici et al 2018; Mehdi et al 2022; Raza et al 2023), and alignment (Mohammed et al 2023). Furthermore, these technologies aid in injury prevention by detecting biomechanical irregularities and high-risk movements, thereby reducing long-term damage and enabling evidence-based rehabilitation protocols. Pose estimation technologies also play a pivotal role in sports broadcasting by delivering real-time visual analytics—such as motion trails and heatmaps—that illuminate player movements and strategies, enhancing viewer engagement and democratizing access to expert-level game analysis (Guo et al 2024; Liu et al 2025; Baumgartner et al 2023; An 2024). These capabilities enrich the overall spectator experience and support a broader audience in understanding complex in-game dynamics. Through the integration of machine learning and artificial intelligence, pose estimation systems now offer accurate, real-time feedback even in dynamic and unconstrained sports environments (Moran et al 2022; Yeung et al 2024; Riccio 2024; Ajay et al 2023). As the demand for precision and data-driven insights grows, pose estimation has emerged as a cornerstone technology—bridging traditional coaching methodologies with advanced performance analytics.

Pose estimation and tracking, the process of identifying and analyzing human body positions and movements, has witnessed remarkable advancements in recent years, driven by innovations in computer vision and artificial intelligence (Ngali and An 2023; Sharma et al 2023; Chamola et al 2024; Chen et al 2020a; Algabri et al 2024; Roggio et al 2024). In the context of sports, these technologies offer transformative potential by providing precise insights into athletes' biomechanics, facilitating performance optimization, and enhancing the overall sports experience (Sharma 2024; Mundt et al 2022; Badiola-Bengoa and Mendez-Zorrilla 2021; Stenum et al 2021a). Pose estimation is the computational process of detecting and predicting the spatial locations of specific key points (e.g., joints, body landmarks) on a human body within an image (Maji et al 2022; Zimmermann et al 2018; Zheng et al 2023; Dang et al 2019; Lan et al 2022; Fang and López 2019) or video (Kocabas et al 2020; Zhou et al 2023; Zheng et al 2023; Dang et al 2024). Pose estimation involves identifying the positions and relationships of various body parts, such as the head, shoulders, elbows, knees, and ankles, to reconstruct a representation of the person's posture or movement (Sigal 2021; Kulkarni et al 2022). Pose estimation can be broadly categorized into two types: 2D (Cao et al 2017; Girdhar et al 2018; Papandreou et al 2017; Liu et al 2022b) and 3D (Liu et al 2022b, 2020c; Zimmermann et al 2018; Moon et al 2018; Gkioxari et al 2014). 2D pose estimation predicts the coordinates of key points in a two-dimensional plane (x, y) based on visual data from a single camera or image. In contrast, 3D pose estimation extends the task into three dimensions (x, y, z), incorporating depth information to capture the body's position and orientation in space. The primary objectives of pose estimation are to interpret human body movement and understand biomechanics, enabling its application in fields like sports (Xi et al 2024; Baumgartner and Klatt 2023; Badiola-Bengoa and Mendez-Zorrilla 2021; Arif et al 2022; Bridgeman et al 2019; Yeung et al 2024), healthcare (Chen et al 2018; Martini et al 2022; Mohammed et al 2023; Raza et al 2023; Shrestha et al 2024; Tluli and Al-Maadeed 2024; Roggio et al 2024), robotics (Yuanwei et al 2024; Deng et al 2020; Bauer et al 2024; Wang et al 2023; Lambrecht 2019), and animation (Kumarapu

and Mukherjee 2021; Tran et al 2013; Amrutha et al 2021; Bridgeman et al 2019). Pose Tracking, on the other hand, builds upon pose estimation by maintaining temporal consistency across sequential frames in a video. While pose estimation analyzes static postures, pose tracking focuses on continuous motion, identifying and following the trajectory of key points over time. This dynamic aspect makes it particularly useful for applications involving motion analysis, such as sports performance tracking and real-time feedback systems. Pose tracking involves challenges such as occlusions (body parts being obstructed) (Zhong et al 2020b, a; Gamez et al 2021; Zheng et al 2022b; Liu et al 2023a; Zou et al 2023; Kim et al 2023; Guidolin et al 2024; Jiang et al 2024; Sun et al 2024a; Cheng et al 2019; Zhang et al 2021a), variable lighting conditions (Crescitelli et al 2020; Lee et al 2023; Liu et al 2023a; Chang et al 2004; Liu et al 2002; Kramer et al 2022; Ai et al 2024; Zhang et al 2023b; Wang et al 2024c), and maintaining consistency even when athletes or objects move rapidly or unpredictably (Bao et al 2020; Wu et al 2022b; Wang et al 2022; Jiang et al 2024; Liu et al 2024c). Advanced approaches integrate temporal models, like Recurrent Neural Networks (RNNs) or Transformers, to achieve robust performance in such scenarios (Chaoyi et al 2021; Xu et al 2022; Zheng et al 2022b; Jin et al 2022; Zhang and Cui 2022; Zou et al 2023; Liu et al 2023a; Sun et al 2024a). Key components of pose estimation and tracking include the detection of key points, which involves identifying anatomical landmarks such as joints or limbs (Sigal 2021; Kulkarni et al 2022). Another critical aspect is feature representation, where meaningful visual or sensor data is extracted to effectively model the movements of body parts. For pose tracking, temporal consistency plays a vital role in ensuring smooth transitions and the continuous tracking of movements across sequential video frames (Li et al 2019; Andriluka et al 2018; Liu et al 2018). Additionally, the integration of depth information (Jun et al 2021; Zhu et al 2008; Marin-Jimenez et al 2018; Wang et al 2016; Ruget et al 2021) or multi-modal data (Lin et al 2023; Deng 2023; Hoang et al 2024; Zhuoins et al 2023), such as depth maps, wearable sensors, or multi-camera systems, enhances the accuracy and reliability of pose estimation and tracking systems.

Recent technological advancements, particularly in artificial intelligence (AI) and machine learning, have significantly enhanced the capabilities of pose estimation systems, making them more accurate, efficient, and adaptable. Deep learning models, such as convolutional neural networks (CNNs) and transformers, have revolutionized the field by enabling the detection and tracking of complex human movements with unprecedented precision (Chen et al 2022; Wu et al 2022a; Xu et al 2022; Zhang and Cui 2022; Fu et al 2023; Yang et al 2023; Bera et al 2023; Liu et al 2023a; Liang et al 2024; Chen et al 2024; Xi et al 2024; Sun et al 2024a). These models excel at handling challenges such as occlusions, varying lighting conditions, and dynamic backgrounds, which are commonly encountered in sports environments (Cheng et al 2019; Zhang et al 2023b; Liu et al 2019; Yang et al 2021; Samkari et al 2023). The integration of advanced algorithms with sensor technologies, such as depth cameras (Jun et al 2021), LiDAR (Ren et al 2024; Zhao et al 2024; Wu et al 2024), and wearable devices (Baldi et al 2019; Yang et al 2020; Mollyn et al 2023; Yu et al 2024b), has further expanded the scope of pose estimation. These tools provide richer data modalities, such as 3D spatial information and real-time feedback, enabling a deeper understanding of biomechanics and movement patterns. Moreover, the development of real-time systems powered by GPU acceleration and optimized neural networks has made it feasible to analyze fast-paced sports activities without compromising accuracy (Voikov et al 2020). These technological breakthroughs have not only improved the reliability of

pose estimation systems but have also opened new avenues for their application. From automated performance analysis to personalized training programs and enhanced broadcasting features, the synergy between AI and machine learning continues to redefine how human motion is studied and utilized in sports science (Frangoudes et al 2022; Singh et al 2023; Yu et al 2024b).

Despite significant advancements in pose estimation and tracking technologies, several gaps in existing research and emerging needs in the field remain unaddressed. Current systems often face limitations in handling real-world complexities, such as occlusions, dynamic sports environments, and variations in athlete demographics and movement patterns (Bridgeman et al 2019; Devi et al 2024; Jafarzadeh et al 2021; Ingwersen et al 2023). These challenges reduce the generalizability and robustness of pose estimation models across diverse sports disciplines, particularly in scenarios involving rapid, high-intensity actions or intricate motions, such as those seen in gymnastics or martial arts. Furthermore, while many studies focus on specific sports or controlled environments, there is a pressing need for systems that can adapt to the diverse contexts and dynamic conditions encountered in live sports settings. For instance, models trained on general-purpose datasets often fail to account for the unique biomechanical nuances of individual or team sports. Additionally, there is a scarcity of annotated datasets specific to various sports disciplines, which hampers the development of more specialized and accurate pose estimation systems. Emerging needs also include the integration of multi-modal data sources, such as wearable sensors and biomechanical measurements, to provide a more holistic understanding of athletic performance. Real-time applications, such as live feedback for athletes and augmented broadcasting experiences, demand lightweight yet accurate models capable of operating under constrained computational resources. Addressing these gaps and needs is crucial for advancing pose estimation technology and maximizing its potential impact in sports science and beyond.

The motivation to study pose estimation in sports stems from its myriad applications. From improving athletic performance through detailed motion analysis to reducing injury risks by detecting biomechanical irregularities, pose estimation has become indispensable. Moreover, in broadcasting, real-time player tracking enriches viewer engagement by overlaying performance metrics and visualizations. Despite these advancements, several challenges persist, such as occlusions during high-intensity sports or variations in environmental conditions, which hinder the deployment of robust systems across diverse sports scenarios. This paper aims to address these issues by presenting a comprehensive survey of pose estimation and tracking methodologies tailored to sports. We examine traditional and modern approaches, highlight the challenges in practical applications, and explore future research directions to address the existing gaps. Unlike previous surveys, this work emphasizes a sport-specific perspective, addressing both team and individual sports contexts, and evaluates recent developments in wearable technologies and machine learning models. By providing an in-depth exploration of methodologies, challenges, and potential innovations, this paper aspires to contribute to the growing body of research on pose estimation and tracking, offering valuable insights for academics, industry practitioners, and sports enthusiasts alike.

Figure 1 shows the taxonomy of this survey, organizing the key research areas in pose estimation and tracking in sports into a structured hierarchy. It categorizes methodologies into traditional and deep learning-based approaches, highlighting various techniques such as feature representation, model types, and neural network architectures. Figure 1 also outlines

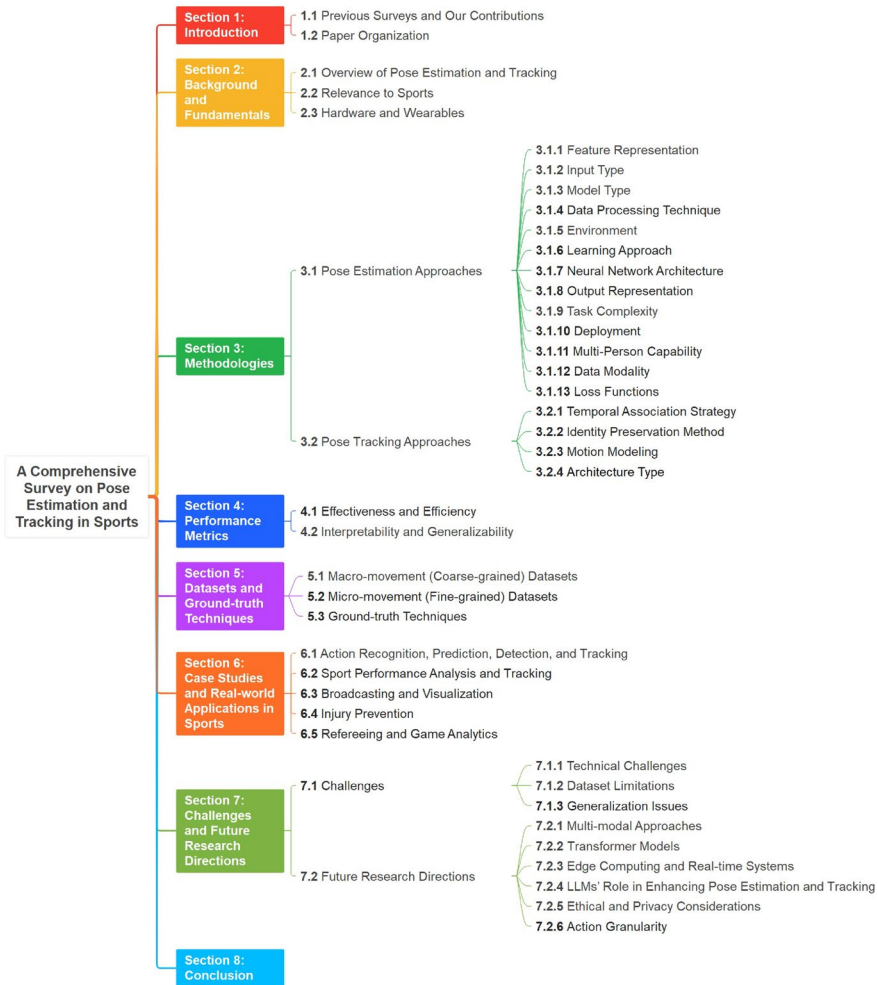


Fig. 1 Hierarchical Taxonomy of Pose Estimation and Tracking in Sports

datasets and evaluation metrics, emphasizing the distinction between macro-movement and micro-movement datasets. Furthermore, it illustrates applications in sports, including performance analysis, injury prevention, and game analytics. Finally, it presents challenges and future research directions, addressing issues like occlusions, dataset limitations, and real-time processing while suggesting advancements through multi-modal approaches, transformers, and edge computing.

1.1 Previous surveys and our contributions

Several recent surveys have reviewed the advancements in pose estimation and tracking technologies, particularly focusing on their applications in sports. Barris and Button (2008) offered a historical perspective on motion tracking, transitioning from manual notation to

automated vision-based systems. Their review highlighted the challenges of applying these technologies in sports, including issues with occlusions, motion complexity, and lighting variations in both outdoor and indoor environments. Badiola-Bengoa and Mendez-Zorrilla (2021) conducted a systematic review on camera-based human pose estimation in sports and physical exercise. Their work focused on the adaptation of general-purpose systems, such as OpenPose, to sports-specific scenarios. The study noted that while general-purpose systems perform well in standard contexts, they often require customization for complex movements and occlusions unique to sports like gymnastics or yoga. Naik et al (2022) reviewed sports video analysis, concentrating on player and ball tracking, event recognition, and AI applications. This survey also discussed the integration of GPU-based platforms for real-time analysis, providing insights into the practical deployment of computer vision systems. The use of embedded platforms and AI algorithms for sports analytics was a key focus. Liu et al (2023b) provided a comprehensive examination of wearable sensor and vision-based technologies in precision sports. This study detailed the methodologies, applications, and challenges of technologies such as GPS, IMU, and vision-based systems for player tracking and performance analysis. Additionally, it discussed data privacy and ethical issues, which are increasingly relevant in the era of big data. Mendes-Neves et al (2023) explored the use of advanced computer vision techniques, including object detection and pose estimation, in sports analytics. They highlighted specific use cases, such as soccer shot speed estimation, showcasing the potential of pose data for improving player performance. The replicability of the methods discussed was emphasized, underlining their practicality for various sports contexts.

Despite the advancements documented in these surveys, there are certain shortcomings and limitations in the comprehensive evaluation and analysis of pose estimation and tracking technologies in sports. While several previous surveys, such as those by (Naik et al 2022) and (Badiola-Bengoa and Mendez-Zorrilla 2021), have focused specifically on pose estimation applications in sports contexts, this review expands upon those efforts by offering a broader and more updated perspective. In contrast to earlier works, our survey comprehensively integrates traditional approaches, deep learning-based methods—including recent advancements such as transformer architectures—and provides a structured discussion of domain-specific challenges and emerging trends. Furthermore, we propose a forward-looking research roadmap, distinguishing this study as a more holistic and strategic resource for researchers and practitioners in sports-related pose estimation. It highlights the limitations of applying generalized systems in sports, such as their struggles with dynamic movements, occlusions, and varying environmental conditions, and provides targeted solutions. In addition, this survey goes beyond merely discussing existing technologies and datasets. It emphasizes the importance of developing sports-specific datasets and explores the potential of multimodal approaches, such as combining RGB cameras, depth sensors, and inertial measurement units (IMUs), to improve system robustness and accuracy. Unlike prior studies that might focus on a single modality or technique, this survey integrates both traditional methods and cutting-edge deep learning models like CNNs, transformers, and hybrid architectures, providing a broad yet focused perspective on advancements in the field. Furthermore, this survey uniquely addresses critical aspects like ethical considerations and computational optimizations, which are essential for deploying pose estimation systems effectively in real-time sports environments. By covering these areas, it offers a comprehensive view that not only enhances theoretical understanding but also provides practical

guidance for implementing these technologies responsibly and efficiently. This approach ensures that the findings are both academically rigorous and directly applicable to real-world challenges in sports analytics.

The comparison of previous surveys and this survey, as outlined in Table 1, highlights the unique contributions of this work. While earlier studies focused on historical perspectives, general-purpose systems, or isolated advancements such as real-time analysis or wearable sensor integration, this survey provides a comprehensive approach. It addresses critical sports-specific challenges, explores multimodal data integration, and emphasizes ethical considerations and computational optimizations.

While Table 1 provides a structural overview of existing surveys in the domain of pose estimation and tracking, it is equally important to articulate the methodological distinctions that differentiate the present work. Prior surveys have often focused either on general-purpose pose estimation systems or have narrowly addressed specific stages of the processing pipeline. In contrast, this survey offers a comprehensive and integrated perspective by systematically analyzing both pose estimation and tracking methodologies, covering traditional approaches as well as recent advancements in deep learning. Notably, we incorporate discussions on transformer-based architectures, hybrid learning strategies, and multimodal data fusion—elements that are largely absent in previous literature. Furthermore, this review is uniquely tailored to the demands of sports-specific applications, emphasizing domain-dependent challenges such as dynamic environments, real-time constraints, and performance variability. Finally, the paper contributes a structured roadmap for future research, which aims to guide the development of more robust and context-aware pose estimation systems in athletic settings—an aspect that is typically underexplored in earlier works.

1.2 Paper organization

The remainder of this survey is structured as follows: Section 2 provides the background and fundamental concepts, including an overview of pose estimation and tracking techniques, and their relevance to sports. Section 3 delves into the methodologies on pose estimation and tracking. Section 4 gives performance metrics. Section 5 focuses on datasets and ground-truthing techniques, discussing data collection, annotation strategies, and challenges in creating high-quality datasets for sports-specific pose estimation and tracking. Section

Table 1 Comparison of previous surveys and this survey

Aspect	Barris (2008)	Badiola (2021)	Naik (2022)	Liu (2023)	Mendes (2023)	This Survey
Historical Perspective	✓	×	×	×	×	×
Focus on General-Purpose Systems	×	✓	×	×	×	×
Real-Time Analysis	×	×	✓	×	×	✓
Wearable Sensor Integration	×	×	×	✓	×	✓
Advanced Computer Vision Techniques	×	×	×	×	✓	✓
Emphasis on Sports-Specific Challenges	×	Partial	×	×	×	✓
Multimodal Approaches	×	×	×	×	×	✓
Ethical Considerations	×	×	×	✓	×	✓
Computational Optimizations	×	×	×	×	×	✓
Integration of Traditional and Deep Learning Methods	×	×	×	×	×	✓

6 explores diverse case studies and real-world applications of pose estimation and tracking in sports, ranging from performance analysis to injury prevention and game analytics. Section 7 highlights the challenges faced in this domain, such as occlusions, domain generalization, and computational constraints, and outlines future research directions, including multimodal approaches, ethical considerations, and real-time systems. Finally, Section 8 concludes the survey by summarizing the key findings in this field.

2 Background and fundamentals

2.1 Overview of pose estimation and tracking

Pose estimation is the process of determining the spatial configuration of key body joints or landmarks in an image or video. It focuses on identifying the precise positions of these key points (e.g., elbows, knees, shoulders) in a single frame, either in 2D or 3D space. Pose estimation is typically a frame-by-frame process, meaning it does not inherently consider temporal relationships between frames in a video sequence (Andriluka et al 2018). Pose tracking builds upon pose estimation by linking detected poses across consecutive frames in a video. It aims to maintain the identity of individuals and track the movement of their key points over time. Pose tracking introduces a temporal component, using motion patterns and consistency to predict the progression of poses, even when keypoints are occluded or temporarily lost (Bridgeman et al 2019). Figure 2 shows an example of pose estimation and tracking in sports. Table 2 summarizes distinctions between pose estimation and pose tracking. Pose estimation and tracking are complementary tasks, often used together in sports analytics to gain both static and dynamic insights into athlete performance and movement patterns. Table 3 summarizes the key components of pose estimation and tracking in sports.

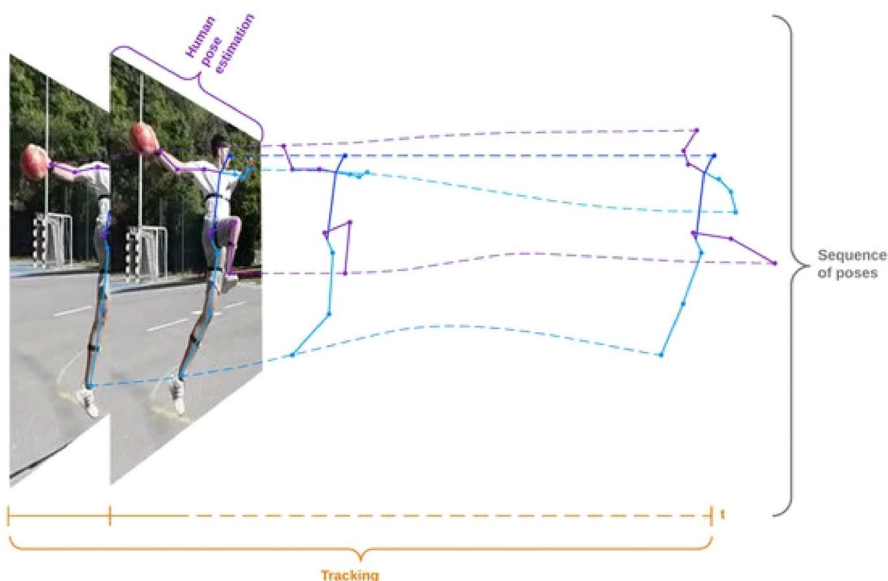


Fig. 2 Pose estimation and tracking in sports (Šajina and Ivašić-Kos 2022)

Table 2 Comparison between pose estimation and pose tracking

Aspect	Pose estimation	Pose tracking
Scope	Single frame analysis.	Continuous analysis across video frames.
Temporal context	No temporal context; independent per frame.	Incorporates temporal relationships between frames.
Output	Keypoints for one frame (static).	Trajectories of keypoints across multiple frames.
Challenges	Accurate keypoint detection in complex poses.	Identity maintenance, occlusions, and smooth transitions.
Techniques	Heatmap regression, CNN-based architectures.	RNNs, optical flow, Kalman filtering, or transformers.
Use cases	Static pose evaluation, snapshot-based assessments.	Motion analysis, athlete tracking, and player identity tracking.

Table 3 Key components of pose estimation and tracking in sports

Key component	Description	Examples
Data acquisition	Collecting input data from cameras, sensors, or other devices.	High-speed cameras for sprinting or swimming strokes.
Preprocessing	Enhancing data quality and preparing it for analysis.	Background subtraction, motion blur correction, and normalization of soccer footage.
Feature representation	Representing body configurations using local or global features.	Heatmaps for joint probabilities, capturing hand or body postures.
Model architectures	Computational models used for pose estimation and tracking.	CNNs for gymnastics poses, RNNs for sequential sports movements.
Temporal analysis	Analyzing motion across frames using temporal relationships.	Tracking a soccer player's sprint trajectory or swimming stroke sequence.
Evaluation metrics	Metrics to assess accuracy, efficiency, and robustness of systems.	PCK, OKS for accuracy; FPS, memory usage for efficiency.
Dataset and ground truth	Datasets with annotated keypoints for training and validation.	Annotated datasets for swimming stroke analysis in various lighting conditions.
Multi-person handling	Tracking multiple individuals in team sports or crowded environments.	Tracking basketball players during a game.
Real-time processing	Ensuring live analysis with minimal latency.	Live feedback on tennis serves or soccer passes during matches.
Application-specific	Tailoring models for specific sports and actions.	Separate models for swimming strokes and soccer dribbles.

2.2 Relevance to sports

Pose estimation and tracking systems are transforming the sports industry by providing unparalleled insights into human movement. These systems enable athletes, coaches, and analysts to examine biomechanics precisely, identifying inefficiencies and optimizing performance (Citraro et al 2020). For example, a runner's stride length and alignment can be analyzed to improve speed and reduce energy wastage. Similarly, a tennis player's serve can be refined by tracking the angles and velocity of arm and body movements. This level of granularity aids in achieving peak performance while minimizing the risk of injury through improved posture and technique.

Injury prevention and rehabilitation are other critical areas where pose estimation systems play a pivotal role. By monitoring athletes' movements in real-time, these systems can detect improper form or biomechanical stress that may lead to injuries. For instance, a

basketball player's jump landing can be analyzed to identify risky knee alignments, helping prevent ligament damage. During rehabilitation, pose tracking ensures that exercises are performed correctly, accelerating recovery and reducing the risk of re-injury (Penichet-Tomas 2024). These tools provide evidence-based approaches to injury management, enhancing athlete safety and career longevity.

Additionally, pose estimation and tracking systems are invaluable in tactical and strategic decision-making, particularly in team sports. They provide detailed insights into player movements, formations, and interactions during gameplay, enabling coaches to develop effective strategies. In soccer, for example, tracking systems can analyze a team's defensive shape or offensive runs, offering real-time adjustments to improve performance (Ashford et al 2021). These systems also enhance the spectator experience by adding data-rich visuals, such as heatmaps or joint trajectories, during live broadcasts, providing fans with deeper insights into the game. Thus, these technologies are not only reshaping athletic performance but also enriching the overall sports ecosystem.

2.3 Hardware and wearables

The application of sensor technologies has facilitated the digitalization of sports activities, enabling the acquisition of precise location and motion tracking data. Wearable sensors include accelerometers, gyroscopes, and biometric tracking devices. They are employed to assess the performance of athletes, monitor and analyze their movements. Furthermore, these sensors are integrated with machine learning algorithms and artificial intelligence methods in motion tracking and recognition processes. They contribute to diverse applications such as preventing sports injuries, personalizing training, and supporting referee decisions. Figure 3 shows the overview of hardware and wearable technologies for sports pose estimation and tracking.

Figure 3 categorizes key technologies into different groups. Camera-based systems include RGB, depth, multi-camera setups, and high-speed cameras for computer vision-based pose estimation. Wearable motion sensors(such as IMUs, EMG, and pressure sensors) provide real-time motion tracking and biomechanical analysis. Wearable tracking devices (such as GPS trackers, smart suits, and motion capture suits) are used for real-time player tracking in various sports. Biomechanical lab equipments(such as optical motion capture systems, force plates, and instrumented treadmills) offer high-precision kinematic and kinetic data for sports science research. AI-powered edge devices enable real-time pose estimation using embedded computing solutions. These technologies collectively enhance performance analysis, injury prevention, and real-time feedback in sports applications. Figure 4 shows some wearable examples associated with sports.

3 Methodologies

3.1 Pose estimation approaches

There are different approaches for pose estimation in the literature. These approaches are categorized based on different criteria, including feature representation, input type, model type, data processing technique, environment, learning approach, neural network archi-

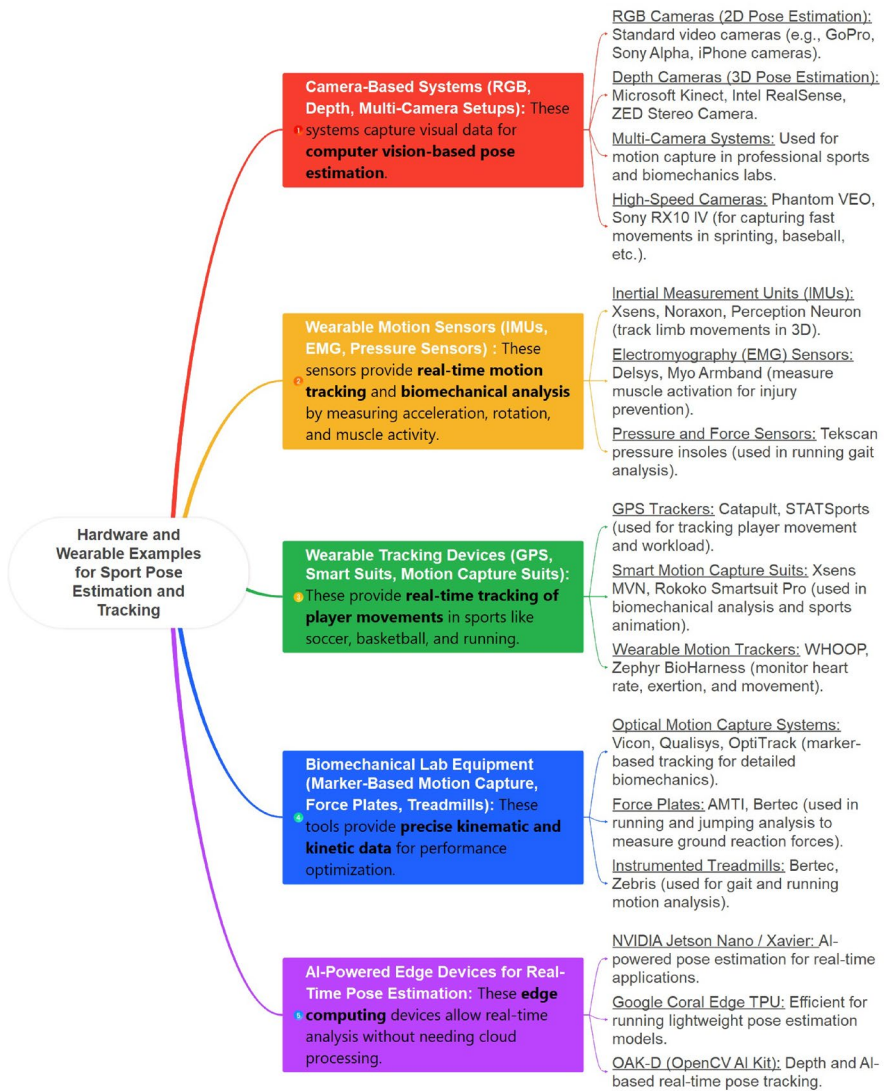


Fig. 3 Hardware and wearable examples for sport pose estimation and tracking

texture, output representation, task complexity, deployment, multi-person capability, data modality, and loss functions. Figure 5 presents a comprehensive categorization of pose estimation methods based on various criteria such as feature representation, input modality, model type, environment, neural network architecture, learning paradigm, and output representation. Feature representation methods include local features, such as HOG, SIFT, and SURF, and global features, such as silhouettes and contours. Based on input type, traditional pose estimation can be performed using either image-based or video-based data. In terms of model type, template-based methods, statistical models like HMM and Kalman filters, and geometric models such as kinematic chains are commonly used. Data processing techniques



Fig. 4 Some wearable devices for sport(<https://www.sportswearable.net/the-hits-misses-and-the-stats-of-the-wearable-world/>)

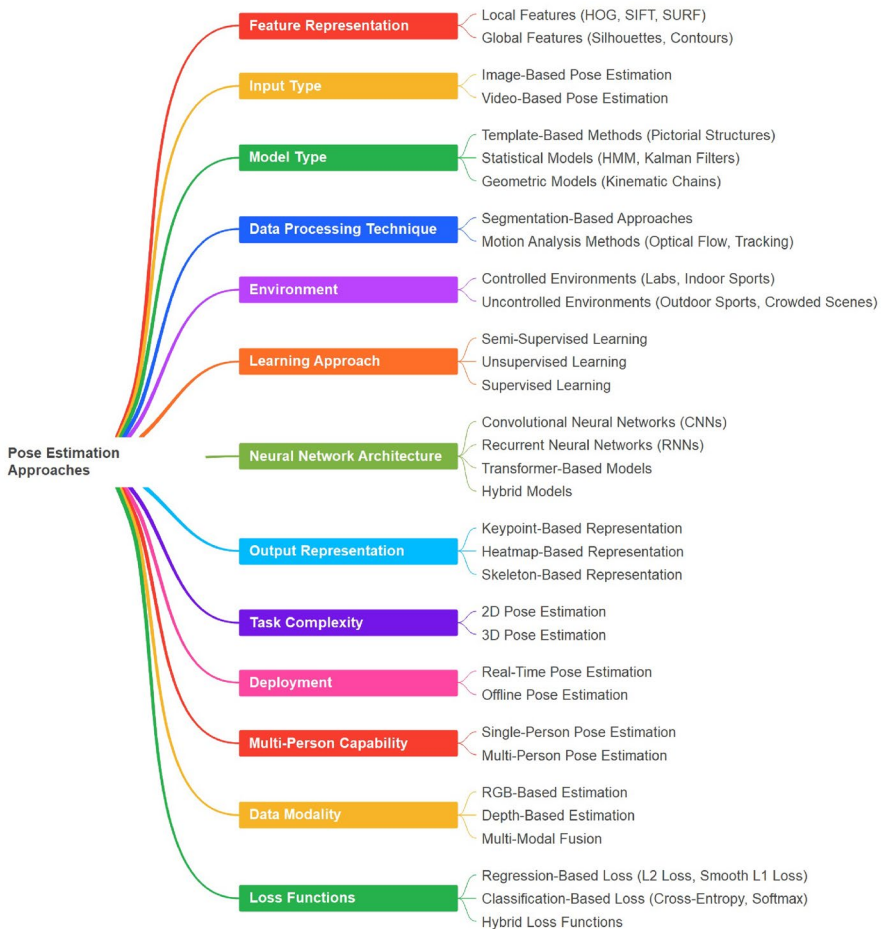


Fig. 5 Classification of pose estimation strategies and techniques

include segmentation-based approaches and motion analysis methods like optical flow and tracking. Finally, pose estimation approaches can be applied in controlled environments, such as laboratories and indoor sports settings, or in uncontrolled environments, including outdoor sports and crowded scenes.

Task complexity is divided into 2D and 3D pose estimation, while deployment type differentiates between real-time and offline processing. Multi-person capability distinguishes between single-person and multi-person pose estimation. Learning approaches include supervised, semi-supervised, and unsupervised learning techniques. Various neural network architectures such as convolutional neural networks (CNNs), recurrent neural networks (RNNs), transformer-based models, and hybrid architectures are commonly used. Data modality is classified into RGB-based estimation, depth-based estimation, and multi-modal fusion techniques. The output representation methods include keypoint-based, heatmap-based, and skeleton-based representations. Finally, different loss functions such as regression-based loss, classification-based loss, and hybrid loss functions are employed to optimize deep learning models for pose estimation.

3.1.1 Feature representation

Feature representation-based approaches consider local or global features. Local features focus on specific points or regions of interest within an image, such as joints or edges, to capture fine-grained information. These features are highly detailed and sensitive to small variations. They are ideal for detecting and tracking precise body movements. In sports, local features are often used to estimate the position of keypoints like elbows, knees, or ankles (Schlegel et al 2024). Techniques such as Histogram of Oriented Gradients (HOG), Scale-Invariant Feature Transform (SIFT), or Speeded-Up Robust Features (SURF) are traditionally employed to extract these features. For example, in a tennis match, local features can be used to track the motion of a player's arm during a serve (Jacquier-Bret and Gorce 2024), identifying the angle at the elbow or wrist (Ramasamy et al 2022). Similarly, in martial arts, local features can help analyze the trajectory of a punch or kick by focusing on specific joint movements. These features are particularly effective when the background is complex or when multiple athletes are present in the frame, as they allow for precise tracking of individual points.

Global features, on the other hand, describe the overall structure or shape of the subject rather than focusing on individual keypoints. These features capture the holistic appearance of the body, often using silhouettes, contours, or shape descriptors (Gościńska and Frejlichowski 2018). They provide a broader perspective and they are useful for analyzing postures or movements where the relationship between multiple joints is essential. In sports, global features are often used to evaluate complex, full-body movements (Nadeem et al 2021). For instance, in gymnastics, global features can analyze a gymnast's posture during a routine, capturing the overall alignment of the body (Díaz-Pereira et al 2014). Similarly, in high jump events, global features can help assess the athlete's arched body shape during the jump (Bao et al 2023). Techniques like silhouette analysis or shape-based matching are commonly employed for such tasks, providing insights into the athlete's form and technique (Vishwakarma and Kapoor 2015).

Local features excel in precision and are suited for analyzing joint-specific movements, such as in running, boxing, or racket sports. Global features offer a macroscopic view, mak-

ing them ideal for assessing overall body posture in activities like gymnastics, yoga, or diving. Both types of features are often combined to provide a comprehensive analysis of sports movements, allowing for detailed biomechanical studies and performance improvement. Figure 6 shows the local and global features in pose estimation.

3.1.2 Input type

Input type-based approaches consider a single image or video (spatiotemporal) as input. Image-based pose estimation focuses on analyzing a single frame to estimate the positions of key body joints and body posture (Han et al 2022). Traditional methods rely on extracting features such as edges, contours, or localized keypoints from an individual image to estimate the athlete's pose. For example, in golf, image-based pose estimation can analyze a snapshot of a golfer's stance to evaluate posture, alignment, and joint angles (Kim et al 2020; Park et al 2017). Similarly, in archery, a single-frame analysis can assess the archer's posture and the positioning of the bow and arrow to provide feedback on technique (Phang et al 2023). Image-based methods are most effective in static or near-static sports activities where the body posture in a single frame can provide meaningful insights.

Video-based pose estimation extends the analysis to sequences of frames, allowing the estimation of body movement and joint trajectories over time (Von Marcard et al 2016; Hayashi et al 2013). Traditional approaches often rely on tracking algorithms such as optical flow, Lucas-Kanade tracking, or Kalman filters to follow the motion of keypoints across multiple frames. This temporal dimension enables the study of dynamic sports activities. For instance, in sprinting, video-based pose estimation can track the runner's limb movements and stride cycle to analyze efficiency and speed (Garrido-Lopez et al 2024). In basketball, it can monitor a player's dribbling or shooting motion to identify areas for improvement (Liu et al 2022a). Video-based methods are particularly useful in sports involving continuous motion, as they capture transitions and sequences that single-frame analysis cannot. Combining image and video-based approaches can offer a robust solution.

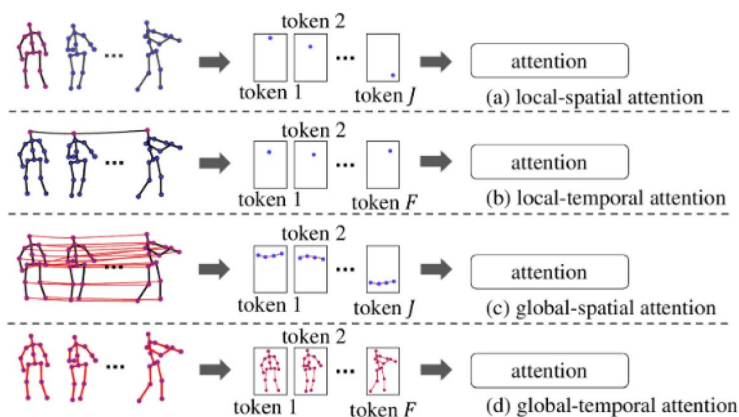


Fig. 6 Local and global features for various types of attentions in pose estimation (No and Lee 2025).

3.1.3 Model type

Model type-based approaches include template-based, statistical models, and geometric models. Template-based methods use predefined templates or models of human body poses to estimate the position of an athlete's joints and limbs. These templates represent typical poses or configurations of body parts and are matched against the input data using similarity measures (Margarito et al 2015). Techniques like the Pictorial Structures Model (Pishchulin et al 2013) or predefined skeletal representations are commonly employed. For example, in yoga, template-based methods can compare an athlete's posture to an ideal pose to provide feedback on alignment and technique. In gymnastics, these methods can assess a gymnast's form during specific movements, such as a handstand or split, by matching the observed pose with stored templates. Template-based approaches are effective in controlled environments but may struggle with unseen or unconventional poses in dynamic sports.

Statistical models rely on probabilistic techniques to estimate poses by modeling the likelihood of body part configurations based on observed data. These methods often use techniques like Hidden Markov Models (HMMs) (Nadeem et al 2021) or Kalman filters to track body movements and predict joint positions over time. Statistical models are particularly useful for dealing with noise or uncertainty in the data. In sports like sprinting, statistical models can track the runner's joint angles and predict their motion trajectory, helping coaches analyze stride patterns (Liu et al 2022a). Similarly, in cycling, these models can monitor the cyclist's pedaling motion, accounting for variability in joint positions due to speed or terrain. Statistical models excel in analyzing repetitive or predictable movements and can adapt to slight variations in body dynamics.

Geometric models use mathematical representations of the human body as a set of rigid segments connected by joints. These models often employ kinematic chains, stick figures, or joint angle constraints to estimate poses. Geometric relationships, such as distances and angles between keypoints, play a critical role in these methods (Rhodin et al 2018). For example, in weightlifting, geometric models can analyze the alignment of the athlete's spine, knees, and hips to ensure proper lifting technique. In basketball, these models can help assess shooting form by evaluating the geometric relationships between the shoulder, elbow, and wrist during the motion. Geometric models are highly interpretable and suitable for tasks where the relationships between body parts are critical to performance evaluation.

3.1.4 Data processing technique

According to data processing techniques, pose estimation is divided into segmentation-based and motion analysis. Segmentation-based approaches involve isolating the athlete from the background and identifying the regions of interest for pose estimation. These methods use techniques such as background subtraction, thresholding, or region-growing algorithms to segment the body or body parts from the surrounding environment (de Souza Vicente et al 2016). Once segmented, the shape and position of the athlete are analyzed to estimate their pose. For example, in swimming, segmentation-based methods can isolate the swimmer's body from the water background to study stroke techniques and body alignment. Segmentation-based approaches work best in controlled environments with consistent lighting and minimal background noise (Sawahata et al 2024).

Motion analysis focuses on tracking the movement of key body parts over time. Techniques such as optical flow, Lucas-Kanade tracking, or motion vector estimation are commonly used to capture the displacement and trajectory of joints or limbs across video frames (Wang 2021). Motion analysis helps understand the dynamics of an athlete's movements rather than static postures. For instance, in basketball, motion analysis can track the player's dribbling motion to evaluate hand-eye coordination and ball control (Battelini 2020). In soccer, it can monitor a player's running pattern and kicking motion to improve agility and shooting technique (Kagalagomb and Dixit 2021). These methods are particularly effective in analyzing sports that involve dynamic and continuous movements. Segmentation-based pose estimation provides detailed static or frame-specific information about the athlete's posture and body shape. Motion analysis-based pose estimation captures the temporal evolution of movements, making it ideal for dynamic sports.

3.1.5 Environment

Pose estimation based on the environment can be controlled or uncontrolled. Controlled environments are scenarios where external factors like lighting, background, and camera placement are carefully regulated. Traditional pose estimation methods thrive in such settings because they can rely on clean data without the complexities introduced by unpredictable surroundings. Techniques like background subtraction, silhouette analysis, or template matching are commonly used (Erol et al 2007). For example, a controlled environment with uniform lighting and a static background allows precise tracking of joint angles and body posture to ensure proper technique in weightlifting. Similarly, controlled environments are used to analyze sprinting or jumping motions under ideal conditions in laboratory-based sports biomechanics studies. These setups are particularly suited for training or rehabilitation scenarios where precision is critical.

Uncontrolled environments are more challenging due to varying lighting, cluttered backgrounds, occlusions, and unpredictable athlete movements (Germann et al 2011). Robust feature extraction techniques like optical flow, HOG, or motion tracking handle these issues (Hofmann and Gavrila 2012). For instance, in a football game, pose estimation must deal with dynamic backgrounds, multiple players, and overlapping movements. Feature-based tracking techniques are used to follow individual players' motions despite these challenges. Similarly, pose estimation needs to account for varying terrain, crowds, and changing light conditions to analyze runners' stride patterns in marathon running. While traditional methods can handle some aspects of these environments, they often require careful pre-processing and may struggle with high variability.

Controlled environments allow for greater precision and simpler processing, making them ideal for sports like gymnastics, swimming, or weightlifting, where fine-grained analysis is crucial. Uncontrolled environments are more realistic but pose significant challenges, making them suitable for sports like football, basketball, or outdoor running, where dynamic conditions are inevitable. Combining techniques optimized for both environments can enhance the versatility of pose estimation systems, enabling robust performance analysis across diverse sports scenarios. Figure 7 shows the environmental challenges with using pose estimation algorithms.

3.1.6 Learning approach

Learning-based approaches can be grouped into three categories. Supervised learning approaches rely on labeled datasets with annotated keypoints to train models for pose estimation. The training process involves providing the model with input images or videos and their corresponding ground truth annotations, allowing it to learn mappings between the input data and joint positions. Popular architectures like OpenPose (Cao et al 2017), HRNet (Sun et al 2019), and DeepLabCut (Mathis et al 2018) are often employed for supervised pose estimation tasks. For instance supervised learning can be used to train a model on annotated datasets of players performing various actions like running, kicking, or dribbling in soccer. This allows the model to accurately estimate poses during gameplay, helping coaches analyze player movements and strategies.

Semi-supervised learning reduces the dependency on large-scale labeled datasets by utilizing a combination of labeled and unlabeled data. These approaches leverage self-supervised pre-training, consistency regularization, or pseudo-labeling to extract meaningful representations from unlabeled data while fine-tuning the model with the available labeled data (Pavlo et al 2019). For example, in tennis, semi-supervised learning can be applied by using a small dataset of annotated images of players' movements along with a larger dataset of unlabeled match footage. This method can train models to predict poses even for unseen movements, enabling performance analysis with minimal manual annotation effort.

Unsupervised learning eliminates the need for labeled datasets by relying on intrinsic patterns in the data (Suzuki et al 2024). These methods use self-supervised techniques, where the model generates pseudo-tasks (e.g., reconstructing input data or predicting missing joints) to learn features. While unsupervised approaches are less accurate than supervised methods, they are valuable in scenarios where annotations are unavailable or infeasible (Rhodin et al 2018). For example, unsupervised learning can be employed to analyze unique and unpredictable tricks by training models on large volumes of unlabeled footage in free-style skateboarding. These models can learn to identify key movement patterns without requiring manual annotation, providing insights into the fluid and creative motion styles of athletes. Each learning paradigm offers unique benefits and trade-offs, and the choice

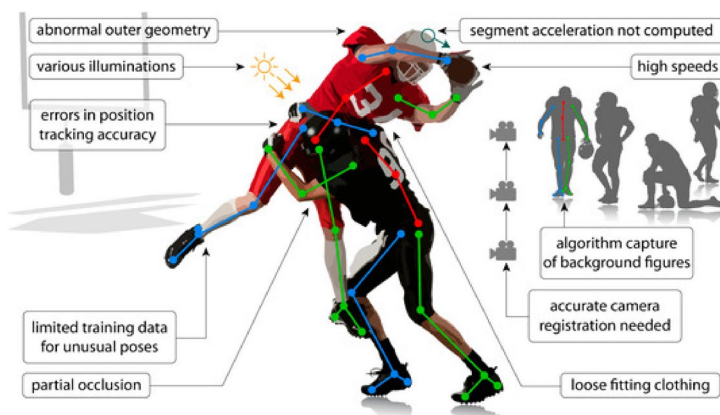


Fig. 7 Environmental challenges with using pose estimation algorithms (Stenum et al 2021a)

depends on the availability of labeled data and the complexity of the sport being analyzed. Figure 8 summarizes the main differences among learning types.

3.1.7 Neural network architecture

According to network architecture, deep learning-based pose estimation may use Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), transformers, and hybrid architectures. CNNs are the cornerstone of many deep learning-based pose estimation models due to their ability to extract spatial features from images. These networks excel at identifying keypoints by learning hierarchical representations of the input data, such as edges, contours, and shapes (Liu and Bhanu 2019). For instance, in basketball, CNN-based models can detect and analyze the player's shooting motion by identifying keypoints like the shoulders, elbows, and wrists. The detailed spatial feature extraction helps in understanding body alignment and movement patterns during shots. RNNs and their variants, such as Long Short-Term Memory (LSTM) networks, are designed to capture temporal dependencies in sequential data (Fok et al 2018). These networks are often employed in sports pose estimation to analyze motion over time, complementing CNNs by handling video-based inputs where temporal continuity is critical. For example, in running analysis, an RNN-based model can track the runner's stride cycle over time, capturing the sequence of joint movements across video frames. This temporal modeling allows for deeper insights into gait patterns and efficiency.

Transformers, initially designed for natural language processing, have recently gained popularity in pose estimation due to their ability to capture long-range dependencies and relationships between keypoints (Ibh et al 2023; AlShami et al 2023). Vision Transformers (ViTs) (Yu et al 2024a) and Transformer-based models like DETection TRansformer (DETR) (Ghosh et al 2022) are now being adapted for pose estimation tasks. For instance, in soccer, a transformer-based model can identify the spatial and temporal relationships between players' keypoints, enabling advanced pose estimation during dynamic gameplay. It can model the interaction between a player's foot and the ball during a kick with high accuracy. Hybrid architectures combine the strengths of different network types, such as CNNs for spatial feature extraction and RNNs or transformers for temporal modeling (Zhao 2023). These architectures provide a comprehensive solution for sports pose estimation by leveraging both spatial and temporal data. For instance, in gymnastics, a hybrid model using CNNs to extract spatial keypoints and RNNs or transformers to analyze the temporal dynamics of

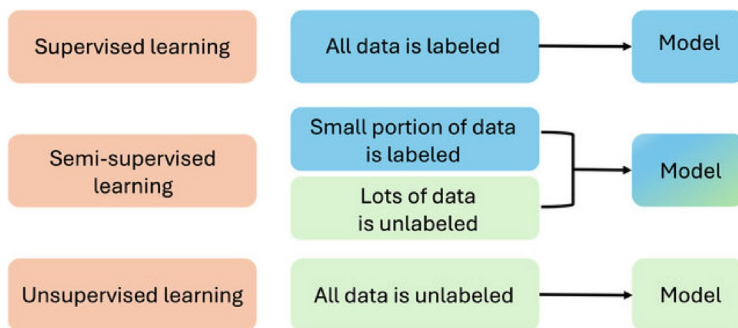


Fig. 8 Learning types for pose estimation

flips or landings can offer detailed insights into the athlete's performance. Such models can assess posture and fluidity across an entire routine. The choice of architecture depends on the nature of the sport and the type of analysis required, with hybrid models often emerging as the most versatile option for holistic performance evaluation. Figure 9 shows the timeline for the mostly used pose estimation models.

3.1.8 Output representation

Output representation can vary among the pose estimation approaches. Keypoint detection directly predicts the coordinates of specific body joints (e.g., wrists, knees, elbows) in an image or video. Deep learning models, such as those using CNNs, are trained to identify and localize these keypoints by outputting their (x, y) coordinates in the image space. This approach is simple and computationally efficient in static or controlled environments, but can struggle with occlusions or complex poses (Nibali et al 2021). For example, keypoint detection can track a player's arm and hand positions during a jump shot in basketball. By focusing on the coordinates of specific joints, this method can provide precise feedback on alignment and movement.

Heatmap-based methods predict a probability map (heatmap) for each keypoint rather than directly outputting coordinates. Each heatmap represents the likelihood of a specific joint being present at each pixel location. This approach allows for more robust and accurate pose estimation, especially in cases of occlusion or noisy input data (Zhou et al 2019). For example, heatmap-based models can analyze the trajectory of a swimmer's arm and leg movements in underwater footage in swimming, even when some body parts are partially obscured. The probabilistic nature of heatmaps makes them resilient to environmental challenges, such as water distortions.

Skeleton-based representation connects detected keypoints to form a skeletal structure that models the body's configuration. This representation provides a higher-level understanding of the athlete's pose and movement by incorporating relationships between joints (Sarwar et al 2023). Skeleton-based methods are often used in conjunction with keypoint detection or heatmaps. For example, a skeleton-based approach can analyze the fighter's posture and movements during a sparring session by connecting joints like shoulders, hips, knees, and ankles in martial arts. The skeletal structure helps in evaluating balance, technique, and power delivery during strikes or defenses.

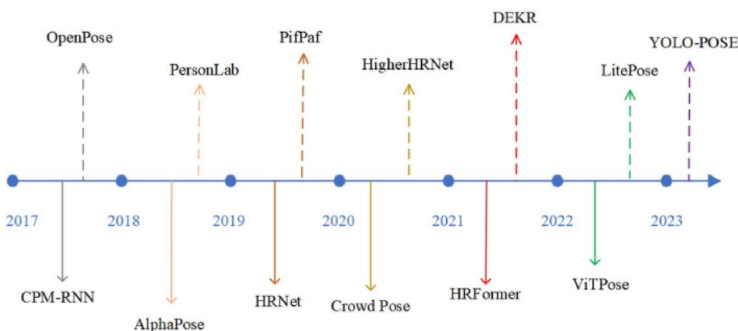


Fig. 9 Timeline for the mostly used pose estimation models (Dong and Du 2024)

3.1.9 Task complexity

Identifying and localizing body joints can be in two-dimensional (2D) or three-dimensional (3D) image space. 2D pose estimation involves identifying and localizing body joints in two-dimensional image space. The output consists of the (x, y) coordinates of keypoints in the image plane, making it computationally simpler and faster than 3D pose estimation (Andriluka et al 2014). While effective, 2D pose estimation may struggle with depth perception, which limits its ability to fully capture complex spatial relationships. 3D pose estimation predicts the three-dimensional coordinates (x, y, z) of body joints, providing a more comprehensive understanding of an athlete's movements in space. It requires additional data, such as depth information or multiple camera views, or uses advanced neural architectures trained on datasets with 3D annotations (Ingwersen et al 2023). Approaches like VideoPose3D (Pavlo et al 2019) or volumetric pose estimation frameworks (Fabbri et al 2020) are popular for 3D tasks. For instance, in gymnastics, 3D pose estimation can analyze a gymnast's mid-air flips and rotations, capturing detailed information about their body orientation and trajectory. By understanding the depth and spatial configuration of joints, 3D estimation provides critical insights for both performance improvement and injury prevention. Figure 10 shows 2D and 3D pose estimation.

In practice, 2D and 3D pose estimation are often used together (Luvizon et al 2018). For instance, 2D pose estimation might be employed for quick, real-time analysis, while 3D estimation is used for in-depth biomechanical studies. This combination allows for a flexible approach tailored to the specific demands of various sports.

3.1.10 Deployment

Deployment refers to the process of making a trained pose estimation model operational for real-world applications in sports. It can be real-time or offline. Real-time pose estimation involves processing input data, such as images or videos, and predicting poses instantaneously or with minimal delay. This requires computationally efficient models that can deliver results in a fraction of a second, making them suitable for applications like live feedback, performance monitoring, or game analysis (Citraro et al 2020). For example, in tennis, real-time pose estimation can track a player's racket position, arm movement, and body posture during a serve. This immediate feedback allows players to adjust their tech-

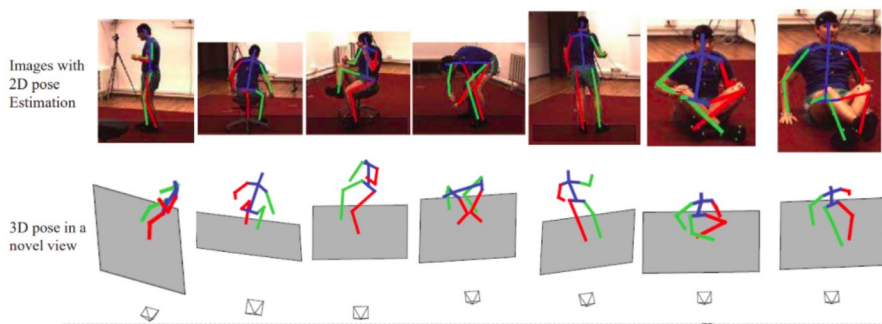


Fig. 10 2D and 3D pose estimation(<https://www.analyticsvidhya.com/blog/2022/04/comprehensive-guide-for-pose-estimation/>)

nique dynamically during training sessions. Real-time systems are also employed in broadcast sports analytics, where pose information can be overlayed on live feeds to enhance viewer experience. Offline pose estimation is conducted after data collection, allowing for the use of more complex models that require significant computational resources. Real-time requirements do not constrain these methods and can leverage larger neural networks, advanced algorithms, and thorough pre-processing for greater accuracy and robustness (Guo et al 2024). For instance, in post-game football analysis, offline methods can evaluate player movements and team dynamics to develop tactics.

3.1.11 Multi-person capability

The number of people in the pose estimation approach is a challenging issue. Single-person pose estimation focuses on detecting and analyzing the pose of one individual in a given input image or video. These models assume that the target athlete is the only person present in the scene or that the model can isolate the individual of interest. Single-person approaches are simpler and faster because they don't require additional mechanisms to identify and separate multiple people in an image (Groos et al 2021; Zhang et al 2021b). For instance, in weightlifting, single-person pose estimation can track an athlete's posture and joint positions to ensure proper form during exercises. By focusing solely on the lifter, the system can provide precise feedback to reduce the risk of injury and improve performance.

Multi-person pose estimation involves detecting and analyzing the poses of multiple individuals in a single frame. This requires additional steps, such as identifying and segmenting individual people before predicting their keypoints (Li et al 2021b). Approaches for multi-person pose estimation typically include top-down and bottom-up methods: Top-down methods detect each person first (e.g., using a bounding box) and then estimate their pose individually. Bottom-Up methods detect all keypoints in the image and group them into poses corresponding to each person. For example, in basketball, multi-person pose estimation can simultaneously track players' movements during gameplay, analyzing team dynamics and strategies. This capability is crucial for understanding interactions between players, such as defensive positioning or offensive patterns (Sajina and Ivacic-Kos 2024). Single-person and multi-person pose estimation systems can complement each other. For example, a soccer training session might use multi-person estimation to analyze team movements during drills and switch to single-person estimation for detailed feedback on an individual player's technique. By leveraging both capabilities, these systems can provide comprehensive insights tailored to different aspects of sports performance.

3.1.12 Data modality

Data modality refers to the type of data or input sources used to perform pose estimation. Different modalities, such as RGB video, depth maps, and multimodal, can be used individually or in combination to address specific challenges in pose estimation. RGB video-based pose estimation relies on standard color images or video footage as input. Deep learning models use features from the RGB channels to detect and localize body keypoints (Mehta et al 2017). This is the most common modality due to the widespread availability of RGB cameras, including smartphones and professional sports cameras. However, RGB-based methods may face challenges in handling occlusions, poor lighting, or dynamic

backgrounds. Depth maps provide information about the distance of each point in the scene from the camera, enabling the model to understand the spatial configuration of the athlete in three dimensions (Leduc et al 2024). Depth data is typically captured using depth sensors like Kinect, LiDAR, or stereo cameras (Guo et al 2024). Depth-based pose estimation is instrumental in addressing challenges like occlusions and complex body orientations, as it includes the z-axis for depth perception. Multimodal approaches combine multiple types of input data, such as RGB video, depth maps, and other sensor data (e.g., inertial measurement units or thermal imaging). By leveraging complementary modalities, these systems can achieve more robust and accurate pose estimation, even in challenging conditions like low lighting, crowded scenes, or occlusions (Wilk et al 2020). Figure 11 shows an example of multimodal pose estimation.

3.1.13 Loss functions

Loss functions play a critical role in training deep learning models by quantifying the difference between predicted and ground truth outputs. In pose estimation for sports, loss functions can be broadly categorized into regression-based and classification-based approaches, each tailored to different output representations and training strategies. Regression-based loss functions are used when the model outputs continuous values, such as the precise coordinates of body keypoints. The loss measures the distance between the predicted keypoint locations and the ground truth coordinates (Li et al 2021a). Classification-based loss functions are used when the output represents probabilities, such as heatmaps, that indicate the likelihood of keypoints being present at specific locations in an image. Each pixel or region of the heatmap is treated as a classification task. Many modern deep learning-based pose estimation systems in sports combine regression and classification-based losses to leverage their respective strengths (Matsune et al 2024). The compositional loss function combines the L1 bone length loss and long-range joint loss (Sárándi et al 2020). Three structure-aware loss (Dabral et al 2018) combines other functions such as illegal-angle loss, symmetry loss, and geometric. Table 4 summarizes the comparison of regression-based and classification-based loss functions in pose estimation.

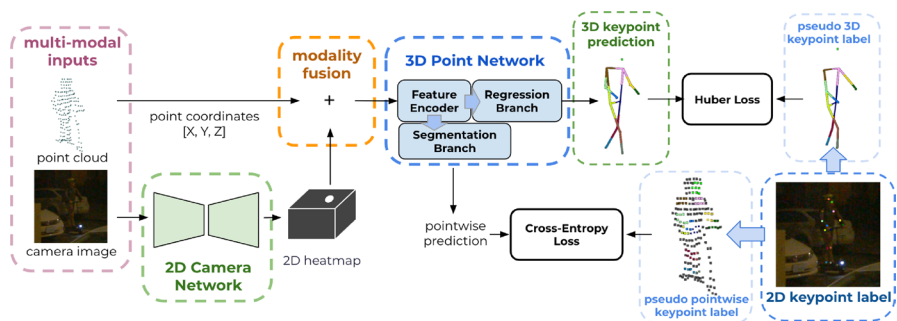


Fig. 11 An example of pose estimation using multimodal data (Zheng et al 2022a)

3.2 Pose tracking approaches

Pose tracking in sports involves identifying and following the spatial positions of athletes' keypoints (e.g., joints, limbs) across sequential video frames (temporal consistency). It builds on pose estimation, which predicts keypoints in a single frame, by adding a temporal component to maintain consistency and continuity across frames (Zecha et al 2019). Pose tracking is crucial in sports for analyzing movements, improving performance, and preventing injuries (Wang et al 2019). There are different pose tracking approaches. Traditional methods rely on handcrafted features and algorithms to track poses. For example, optical flow techniques calculate motion vectors between frames to track keypoints. Traditional pose tracking methods struggle with complex movements, occlusions, or variations in lighting, making them less suitable for dynamic sports scenes.

Modern pose tracking approaches utilize deep learning to enhance accuracy and robustness. These methods integrate pose estimation networks with temporal models, such as RNNs or transformers, to model the temporal relationships between frames (Fan et al 2022; Zhou et al 2023). For example, a deep learning model tracks players' joint movements over time, allowing analysis of dribbling, shooting, and defensive stances. In team sports like soccer or basketball, multi-object pose tracking is essential to monitor multiple players simultaneously (Wojke et al 2017; Du et al 2023). While challenges remain, advancements in deep learning and computational resources continue to push the boundaries of pose tracking, enabling increasingly accurate and efficient systems for sports applications.

Figure 12 provides an overview of the core design elements in pose tracking systems. It emphasizes temporal association strategies, identity preservation methods, motion modeling, and integration of re-identification or multi-frame association techniques. These aspects are critical for maintaining trajectory consistency in dynamic sports environments, especially during occlusions or rapid player movements.

3.2.1 Temporal association strategy

In dynamic sports scenes, pose tracking methods must robustly associate detected poses across consecutive frames, despite challenges like rapid motion and occlusion. Temporal association refers to how algorithms link poses (or people) from frame to frame to form continuous trajectories. Recent works have explored both learning-based and algorithmic strategies for this data association. For example, Rafi et al (2020) introduce a keypoint

Table 4 Comparison of regression-based and classification-based loss functions in pose estimation

Criteria	Regression-based	Classification-based
Output representation	Continuous keypoint coordinates (x, y)	Heatmaps with probabilities for keypoint locations
Loss function examples	MSE, MAE	Cross-Entropy, KL Divergence
Handling occlusions	Limited	Better due to probabilistic outputs
Precision	High	Depends on post-processing resolution
Ideal applications	Controlled environments, solo sports (e.g., tennis, golf)	Crowded scenes, team sports (e.g., soccer, basketball)

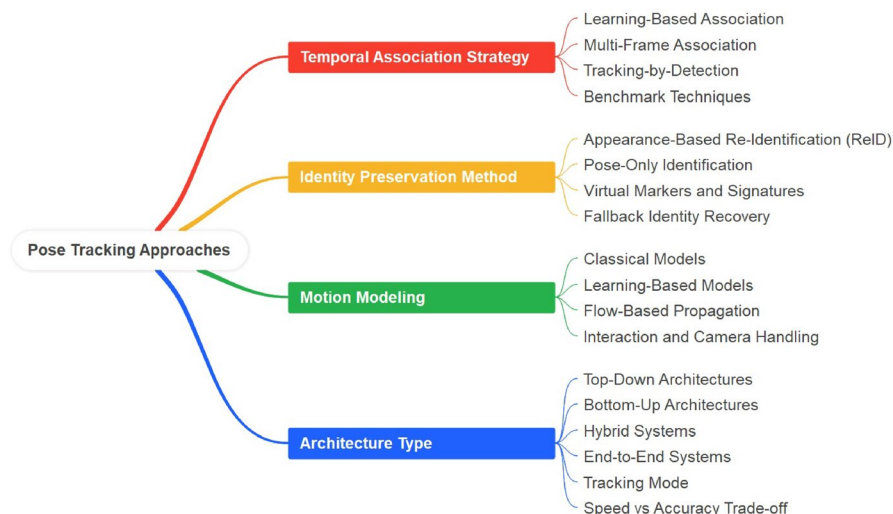


Fig. 12 Classification of pose tracking strategies and architectures

correspondence network that learns affinity maps between poses in consecutive frames, enabling more reliable bipartite matching of poses via the Hungarian algorithm. This approach, known as CorrTrack (Cao et al 2024), significantly improves association accuracy by predicting correspondences even under occlusions or pose changes. Similarly, Snower et al (2020) propose a Pose Entailment model using a Transformer to directly infer whether a pose in one frame corresponds to a pose in another, relying only on keypoint layouts without image appearance. This learned temporal linking achieved state-of-the-art pose tracking on PoseTrack benchmarks while being efficient in sports settings where real-time operation is needed. Other works employ multi-frame association strategies: Fang et al (2024) present a bidirectional temporal pose matching (BTPM) scheme that propagates pose estimates backward and forward in time to handle sudden pose changes, and supplement it with an Association Across Multiple Frames module for long-term tracking. Top-down tracking-by-detection pipelines in sports (e.g. using person detectors per frame) typically associate poses by matching bounding boxes or keypoint similarities across frames. For instance, Liu et al (2024a) use detected poses from each volleyball frame and then apply the DeepSORT tracker to associate them over time. Although simpler, such methods can struggle if players move unpredictably or leave the frame. To address this, some sports tracking systems maintain short track histories and perform delayed association. In the PoseTrack21 (Doering et al 2022) benchmark, an online algorithm is improved by keeping recent tracklets and attempting to reconnect lost poses when they reappear, using stored pose features or re-identification descriptors. Overall, the temporal association strategies have evolved from basic framewise matching to sophisticated learned affinity modeling and multi-frame reasoning, which are crucial for sports videos with frequent player interactions and rapid movements.

3.2.2 Identity preservation method

Pose tracking in multi-player sports requires not only detecting poses but also preserving each player's identity across frames (and sometimes across cameras). An identity preservation method focuses on minimizing identity switches - i.e. ensuring that each trajectory of poses belongs to the same athlete. Traditional multi-object trackers suffered frequent ID switches whenever players overlapped or left the frame (Maglo et al 2022). Appearance-based re-identification (ReID) techniques are a common solution: trackers like CorrTrack (Cao et al 2024; Doering et al 2022) augmented with person ReID features show markedly fewer identity swaps. Doering et al (2022) demonstrated that adding a ReID embedding comparison when linking poses (CorrTrack w. ReID) significantly reduced ID switches on the PoseTrack21 multi-person pose tracking benchmark. Likewise, integrating DeepSORT into pose tracking pipelines provides an identity module - DeepSORT computes a CNN-based appearance vector for each person which, combined with motion, helps maintain consistent IDs across frames. Liu et al (2024a) leverage this in volleyball footage, ensuring each player's pose is tagged with a unique ID over time. Another approach is to use the pose configuration itself as an identity signature. Snower et al (2020) showed that accurate pose layouts alone can preserve identities, by matching poses with a Transformer without any color or texture cues. Their keypoint-only tracker achieved performance on par with appearance-based methods, highlighting that discriminative pose features (e.g. height, limb proportions) can distinguish players even in identical uniforms. For more explicit identity cues, some sports-focused methods introduce unique markers. Yin et al (2024) propose a pose-based virtual marker scheme for basketball, effectively tagging each player with an invisible identifier in the model's input space. This novel method yielded an average High-level tracking accuracy (HOTA) over 72% and zero ID switches in 3v3 basketball games, demonstrating extremely strong identity preservation by design. When such specialized cues are not available, trackers handle identity by robust data association over time (as discussed above) and by reinitializing lost tracks carefully. For example, Fang et al (2024) re-run their matching procedure when association fails, essentially treating that frame as a new keyframe for identity assignment. Maintaining identity is especially challenging in sports where players frequently occlude each other and wear similar jerseys. Integrating person identification mechanisms into pose tracking models is critical for sports analytics applications. Consequently, state-of-the-art sports pose trackers often combine pose estimators with identity modules (appearance networks or pose-specific classifiers) and even offline identity merge steps to correct any fragmentation. This ensures that the output trajectories can be reliably used for performance analysis per individual athlete.

3.2.3 Motion modeling

Effective pose tracking also requires modeling how players move, i.e., a motion model that predicts or smooths the trajectory of the pose between frames. Simple motion models assume near-constant velocity or small inter-frame displacement, which can be handled by linear predictors like Kalman filters. Many sports pose trackers embed a Kalman filter at their core to update each person's predicted position in the next frame, aiding both association and smoothing. (Buizza et al 2020) use Kalman filters to track multiple skeletons in real time, allowing their system to "fill in" pose estimates for intermediate frames and reduce jit-

ter, which improves overall temporal consistency. Similarly, the DeepSORT tracker (used in Liu et al.) applies a Kalman filter per athlete to predict the bounding box (and implicitly the pose location) in the next frame, which is then corrected by the new observation. However, sports movements are often highly non-linear (quick direction changes, jumps, etc.), where basic linear models struggle. To address this, researchers have begun integrating advanced learning-based motion predictors. (Han et al 2025) introduce ETTrack, which replaces the standard filter with a Transformer-based motion model that learns both short-term and long-term motion patterns from data. By capturing complex acceleration profiles, their method better forecasts future positions, yielding improved tracking on dynamic sports sequences (e.g., a 74.4% HOTA on the SportsMOT dataset). Another strategy for motion modeling is the use of optical flow or pose flow to propagate keypoint locations. AlphaPose's earlier tracking module, PoseFlow, propagated poses via optical flow and spatial affinities. While optical flow can quickly predict where joints move in the next frame, it may drift during long occlusions. Thus, some trackers combine flow with re-detection: Fang et al (2024) perform forward pose propagation for a few frames (assuming smooth motion) but if the propagated pose no longer matches the detection, they trigger a re-association by treating that frame as a new keyframe. In multi-person sports scenes, modeling interactions is also important - when players collide or exchange positions, purely individual motion models can fail. Some approaches maintain multiple hypotheses or use a motion confidence check to decide when to trust the model vs. re-initialize a track. Additionally, camera motion (e.g., panning in a soccer broadcast) must be accounted for. Tracktor++ (Bergmann et al 2019) style methods perform frame-to-frame image alignment and then assume constant-velocity motion to update detections in low-frame-rate videos. Such techniques have been adapted to pose tracking by first projecting previous poses according to camera movement before matching. Overall, motion modeling in sports pose tracking has progressed from basic filtering to learning-rich predictors. The inclusion of motion predictors that can adapt to rapid and non-linear movements greatly enhances a tracker's ability to maintain continuous pose trajectories in fast-paced sports scenarios.

3.2.4 Architecture type

Pose tracking approaches can be categorized by their architecture - the design of the overall system and how pose estimation and tracking are combined. One major distinction is top-down vs. bottom-up processing. Top-down architectures first detect each player (often via a person bounding box detector) and then perform single-person pose estimation for each detection. This has the advantage of focusing on each athlete separately, often yielding high accuracy per pose; many sports analytics systems use top-down pipelines with a pretrained detector (e.g. YOLO or Faster R-CNN) followed by a pose network like HRNet for each player (Sun et al 2019). The tracking step in top-down methods can be a separate module that links the detected poses frame to frame (for example, matching or reID as discussed). In contrast, bottom-up architectures detect all keypoints in the frame jointly and group them into persons, inherently handling multiple people without a person detector. OpenPose and its successors (like PifPaf) are classic bottom-up pose estimators; when extended to video, they can produce spatio-temporal affinity fields that connect joints across frames. (Raaj et al 2019)'s STAF network is a bottom-up tracker that learns part affinity fields in time and space, achieving frame-rate tracking invariant to number of people. Bottom-up methods

scale well to many players (important for sports like soccer with 22 players), but can be challenged by occlusions since they lack an explicit notion of per-person identity until after grouping. To combine strengths, some recent systems use hybrid architectures: for example, PoseTrack21 (Doering et al 2022) baselines explored a top-down pose detector with a bottom-up correspondence network for linking poses, CorrTrack (Cao et al 2024).

Another architectural consideration is whether tracking is done in a post-processing step or end-to-end. Traditional approaches perform pose estimation per frame, then separately apply a tracking algorithm to the sequence (two-stage pipeline). Newer approaches aim for end-to-end learning of pose tracking. An example in 2D is KeyTrack (Snower et al 2020), which integrates a pose estimator and a tracking Transformer in one framework - the network outputs refined poses and also learns to predict if consecutive poses belong to the same individual. This unified model greatly streamlines the processing and was able to run in real-time while maintaining high accuracy. In 3D multi-view sports scenarios, end-to-end designs have also emerged: Reddy et al (2021) introduce TesseTrack, a single network that performs multi-camera 3D pose estimation and temporal association simultaneously. Their framework voxelizes input from multiple cameras and produces 3D pose sequences with a differentiable matching layer for tracking, rather than solving each sub-problem independently. This kind of architecture is beneficial in sports settings with dedicated multi-camera setups (e.g., motion capture for training or broadcast systems) and provides robustness to varying camera counts.

We also note the distinction between online vs. offline tracking architectures. Online models process frame-by-frame (suitable for live sports analytics), whereas offline methods may optimize trajectories over the entire sequence. Some pose trackers allow an offline refinement stage - for instance, an initial online tracking is produced, then an offline optimization merges fragmented tracklets or adjusts poses for temporal consistency. Such offline smoothing can significantly reduce errors at the cost of latency, making it a viable option for post-game analysis but not for real-time feedback.

Finally, architecture choices influence the speed/accuracy trade-off. Lightweight models (e.g., EfficientPose with a Temporal CNN as used by some sports action systems) prioritize speed for real-time use, whereas larger models (e.g., using transformers or 3D CNNs across frames) may achieve higher accuracy but require more computation. The state-of-the-art in sports pose tracking often employs a top-down approach with a strong pose backbone (like HRNet or similarly powerful CNN) and a tracking head or algorithm that can run in parallel, achieving a balance of precision and speed. In summary, the architecture of pose trackers ranges from modular two-stage systems to unified end-to-end networks. Current trends favor closer integration of pose estimation and tracking (learning joint features for both) and leveraging temporal context in the model architecture itself, which has proven beneficial for the complex dynamics encountered in sports.

4 Performance metrics

This section gives evaluation metrics for pose estimation and tracking in sports. They are mainly divided into two groups: effectiveness and efficiency-based metrics and interoperability and generalizability-based metrics.

4.1 Effectiveness and efficiency

To evaluate pose estimation and tracking systems, it is essential to measure their effectiveness (accuracy and reliability) and efficiency (speed and resource utilization).

Percentage of Correct Keypoints (PCK) measures the percentage of keypoints detected correctly within a threshold distance from the ground truth. The threshold is often defined as a fraction of a reference length (e.g., torso length). PCK evaluates pose estimation accuracy, particularly in single-frame scenarios.

$$\text{PCK} = \frac{\text{Number of correct keypoints}}{\text{Total number of keypoints}} \times 100$$

Percentage of Correct Parts (PCP) measures the percentage of correctly detected limbs or parts (e.g., arm, leg) based on the proximity of their predicted endpoints to the ground truth endpoints.

$$\text{PCP} = \frac{\text{Number of correctly detected parts}}{\text{Total number of parts}} \times 100$$

Object Keypoint Similarity (OKS) normalizes the correctness of keypoints by their visibility and scale. It is a more refined accuracy metric, used in multi-person pose estimation.

$$\text{OKS} = \frac{\sum_i \exp\left(-\frac{d_i^2}{2s^2k_i^2}\right) \delta(v_i > 0)}{\sum_i \delta(v_i > 0)}$$

where d_i is the distance between the predicted and ground truth keypoints, s is the object scale, k_i is a per-keypoint constant, and v_i indicates keypoint visibility.

Mean Per Joint Position Error (MPJPE) measures the average Euclidean distance between predicted and ground truth keypoints. It is commonly used in 3D pose estimation.

$$\text{MPJPE} = \frac{1}{N} \sum_{i=1}^N \|p_i - g_i\|_2$$

where N is the number of joints, p_i is the predicted position, and g_i is the ground truth position.

Procrustes Aligned Mean Per Joint Position Error (PA-MPJPE) is also known as Reconstruction Error, is a performance metric used in 3D human pose estimation to evaluate how well the predicted 3D poses align with the ground truth poses. It is often used to measure the accuracy of predicted joint positions by accounting for any global translation and rotation differences between the predicted and ground truth poses.

$$\text{PA-MPJPE} = \frac{1}{N} \sum_{i=1}^N \|p_i - g_i\|_2$$

where N is the total number of joints, p_i and g_i are the Procrustes-aligned predicted and ground truth joint positions, respectively.

Average Precision (AP) and **Average Recall (AR)** are commonly used metrics for evaluating object detection and pose estimation models, particularly in terms of their ability to correctly identify keypoints or body parts. These metrics are derived from precision and recall, and they assess the trade-off between false positives and false negatives. AP measures the precision of a model across different levels of recall. It is typically computed by integrating the precision-recall curve.

$$AP = \int_0^1 \text{Precision}(\text{Recall}) d(\text{Recall})$$

AR is the average recall across different intersection-over-union (IoU) thresholds or across different part types (in the context of pose estimation, this could be limbs, joints, or body parts).

$$AR = \frac{1}{N} \sum_{i=1}^N \text{Recall}_i$$

where N is the number of thresholds (e.g., different IoU thresholds or body parts) and Recall_i is recall at threshold i .

Tracking Accuracy (TA) evaluates how well the poses are tracked over time, focusing on consistency across frames.

$$TA = \frac{\text{Number of correct associations across frames}}{\text{Total associations}} \times 100$$

Mean Intersection over Union (mIoU) measures the overlap between the predicted and ground truth segmentation masks or bounding boxes for body parts. It evaluates the accuracy of region-based pose estimation.

$$mIoU = \frac{1}{N} \sum_{i=1}^N \frac{\text{Area of overlap}_i}{\text{Area of union}_i}$$

Frames Per Second (FPS) measures how many frames the model processes per second. Higher FPS indicates better real-time performance. It determines suitability for real-time applications like live sports broadcasting.

$$FPS = \frac{\text{Total processed frames}}{\text{Total time taken (s)}}$$

Inference Time (IT) is the average time taken by the model to process a single frame. Lower inference times indicate better responsiveness.

$$IT = \frac{\text{Total time taken (s)}}{\text{Total processed frames}}$$

Memory Usage (MU) is the amount of memory consumed during model execution. Efficient models use minimal memory. It ensures the model can run on resource-constrained devices like edge computing systems.

$$MU = \text{Peak memory usage (MB/GB)}$$

Energy Consumption (EC) measures the total energy consumed during pose estimation or tracking. It is relevant for battery-powered devices used in wearable sports analytics.

$$EC = \text{Power (W)} \times \text{Time (s)}$$

Accuracy vs. Speed Trade-off (A-STO) balances pose estimation accuracy and processing speed for practical deployment. It highlights models that are both accurate and efficient.

$$A\text{-STO} = \frac{\text{Accuracy metric (e.g., PCK)}}{\text{Inference time (IT)}}$$

Computational Efficiency (CE) measures how efficiently the model utilizes computational resources. It is used to optimize models for deployment on hardware with limited resources.

$$CE = \frac{\text{Frames per second (FPS)}}{\text{Memory usage (MU)}}$$

4.2 Interpretability and generalizability

Interpretability metrics focus on how well the model's predictions can be understood and related to the input data. Generalizability metrics evaluate the model's ability to perform well across diverse conditions, datasets, or environments.

Confidence Score (CS) measures the confidence of the model in detecting and localizing keypoints. Higher confidence scores imply more reliable predictions for specific keypoints.

$$CS_i = \max(H_i(x, y))$$

where $H_i(x, y)$ is heatmap value at position (x, y) for keypoint i .

Attention Map Overlap (AMO) evaluates how well the model's attention regions align with ground truth regions. It helps interpret which areas of the input image influence the model's predictions.

$$AMO = \frac{\text{Area of overlap}}{\text{Area of union}}$$

Cross-Dataset Performance (CDP) measures the model's accuracy when evaluated on a dataset it has not been trained on. Higher CDP indicates better generalization across datasets.

$$\text{CDP} = \frac{\text{Accuracy on unseen dataset}}{\text{Accuracy on training dataset}} \times 100$$

Domain Gap Error (DGE) quantifies the performance degradation when moving from the training domain (source) to a different domain (target).

$$\text{DGE} = \text{Accuracy}_{\text{source}} - \text{Accuracy}_{\text{target}}$$

Robustness to Occlusion (RTO) assesses the model's ability to predict poses accurately in the presence of occlusion. Higher RTO values imply better robustness under challenging conditions.

$$\text{RTO} = \frac{\text{Accuracy with occlusion}}{\text{Accuracy without occlusion}} \times 100$$

Multi-Environment Generalization (MEG) evaluates performance across different environments (e.g., lighting, background complexity).

$$\text{MEG} = \frac{1}{N} \sum_{i=1}^N \text{Accuracy}_i$$

where N is the number of different environments and Accuracy_i is the accuracy in environment i .

5 Datasets and ground-truth techniques

In this section, we present fine-grained and coarse-grained action datasets and ground-truth techniques for obtaining high-quality datasets. Action granularity refers to the level of detail at which athletic movements or actions are analyzed during pose estimation and tracking. In sports, the granularity of action analysis plays a significant role in how effectively these technologies contribute to performance optimization, injury prevention, and decision-making. Higher granularity enables the capture of subtle, intricate movements (such as slight variations in an athlete's hand positioning or foot angle), which are crucial for assessing technique. This level of detail can identify inefficiencies, helping athletes fine-tune their skills. Figure 13 shows coarse and fine-grained action examples.

Figure 14 shows the sports based on the level of detail required for pose estimation and tracking. Sports are divided into three categories: fine-grained, coarse-grained, and hybrid. Fine-grained sports require highly detailed motion analysis, often focusing on precise joint angles, limb movements, or micro-adjustments. Examples include gymnastics, weightlifting, archery, and golf, where small variations in posture or movement significantly impact performance. Coarse-grained sports primarily involve large-scale body movements and

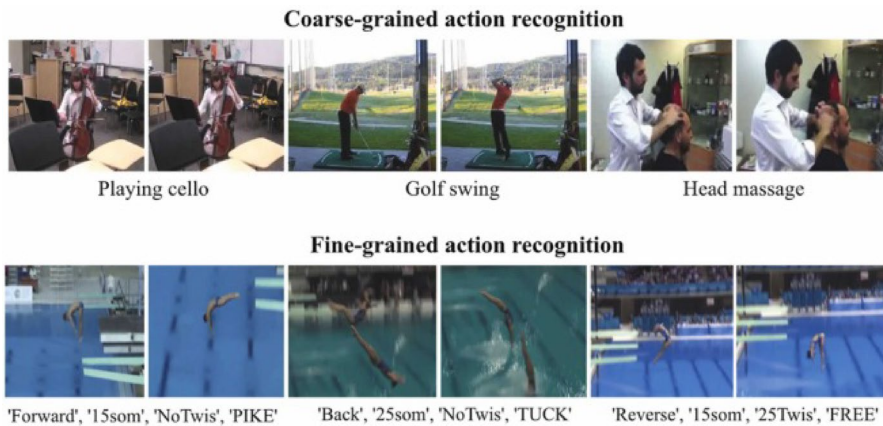


Fig. 13 Coarse-grained and fine-grained action examples (Ma et al 2024)

broader motion patterns, such as team sports (soccer, basketball, American football), track and field, and cycling. These sports emphasize whole-body coordination and positional tracking over detailed joint articulation. Hybrid sports exhibit both fine- and coarse-grained characteristics depending on the context. For instance, in tennis, macro-level court movement is combined with micro-level stroke techniques, while soccer involves both large-scale player movement and detailed ball control. This classification provides insights into the varying demands of pose estimation models across different sports disciplines.

5.1 Macro-action datasets

These sports emphasize larger, full-body movements, focusing on general motion patterns or overall poses (coarse-grained). Precision in fine details is less critical than the efficiency and power of the motion. This subsection presents pose estimation and tracking datasets in coarse-grained sports.

Soccer: In the context of soccer, datasets play a crucial role in pose estimation, motion analysis, and performance evaluation. SoccerNet¹, a prominent dataset, offers annotated match events such as player positions, passes, and goals, making it suitable for pose estimation and event detection. DeepLabCut² is highlighted for its customizable application in motion analysis, supporting both human and animal movement studies. PoseTrack³ is recognized for multi-person tracking and pose estimation in crowded scenes, including sports environments. Broadcast Sports Datasets⁴ and FIFA Analytics Platforms⁵ provide detailed information on player movements, team tactics, and match scenarios, with FIFA datasets focusing on sensor and 3D pose data. The datasets vary in size, from gigabytes to terabytes, and include formats such as MP4 videos, JSON annotations, and proprietary files. Col-

¹ <https://www.soccer-net.org/>

² github.com/DeepLabCut

³ <http://posetrack.net/>

⁴ <https://www.thesports.com/>

⁵ <https://fdp.fifa.org/>

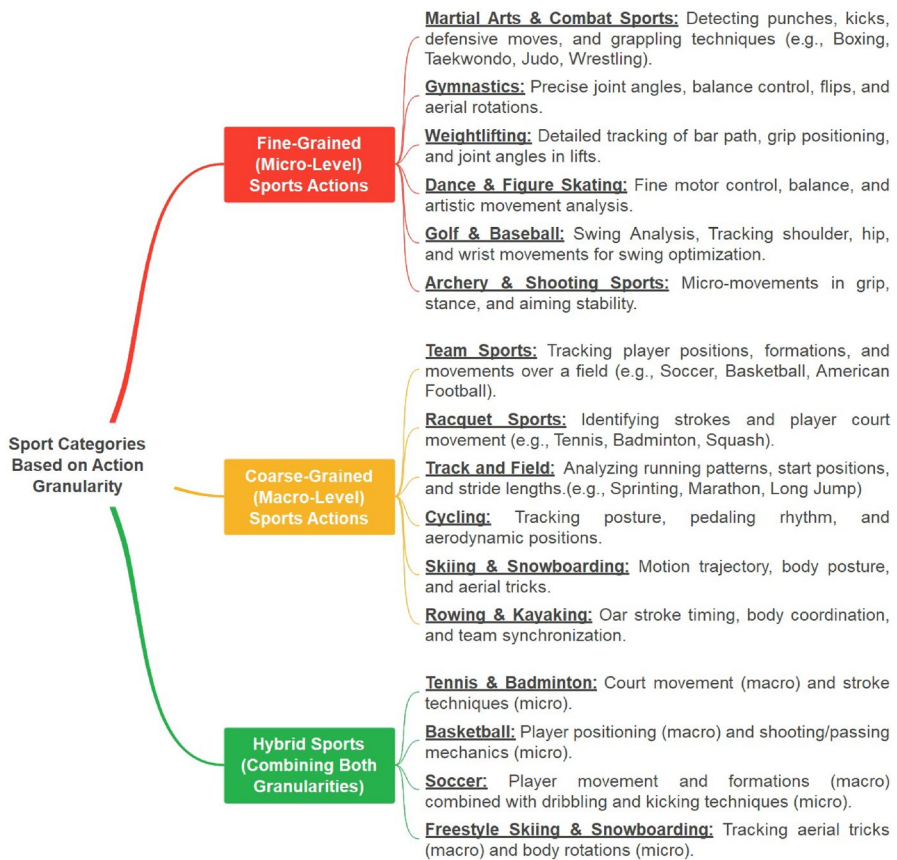


Fig. 14 Sport categories based on action granularity

lectively, these datasets offer a comprehensive foundation for research and applications in soccer analytics.

Basketball: Basketball analytics has greatly benefited from accessible and widely-used datasets tailored for pose estimation, pose tracking, and action recognition. The LiCamPose dataset⁶ combines multi-view LiDAR and RGB cameras for robust 3D human pose estimation and is openly available for research purposes. The NBA2K dataset⁷ is another publicly available resource, widely used for keypoint detection and pose tracking in dynamic basketball scenarios, leveraging multi-camera setups. Additionally, the OpenPose-based Basketball dataset⁸, an extension of the OpenPose framework, focuses on action recognition and detailed pose estimation in sports settings. The NCAA Dataset (Ramanathan et al 2016), focused on basketball action recognition, provides a collection of annotated game footage that supports multi-player pose estimation and action recognition research. The APIDIS (Autonomous Production of Images based on Distributed and Intelligent Sensing) Data-

⁶<https://github.com/Yu-Yy/LiCamPose>

⁷<https://github.com/luyangzhu/NBA2K-dataset>

⁸<https://github.com/OpenPose-Basketball-Dataset>

set (De Vleeschouwer and Delannay 2009) captures multi-camera basketball game footage with detailed annotations, enabling research in multi-person tracking, pose estimation, and team activity analysis.

American Football: In the field of American football, primary datasets such as the OSU Dataset and the Sports-1M Dataset are widely used for pose estimation and tracking for performance analysis and tactical evaluations. The OSU Football Dataset (Hess and Fern 2009), compiled by Hess, consists of annotated American football videos, specifically captured using the “All 22” camera angle, which ensures that all players on the field are visible. This dataset has been widely employed in studies focused on player tracking, field registration, and pose estimation. Similarly, the Sports-1M Dataset (Karpathy et al 2014) comprises over a million YouTube videos categorized into 487 sports classes, including a substantial number of American football-related recordings. In addition to these video-based datasets, pose estimation and tracking datasets that leverage player movement data have been developed to enhance American football analytics. The NFL Big Data Bowl dataset⁹, provided by the National Football League (NFL), includes player tracking data with (x,y) coordinates recorded at various time frames, allowing for in-depth analysis of player movement and game dynamics. Furthermore, custom datasets for offensive formation recognition have been compiled to support automated classification of team formations and tactical evaluations (Atmosukarto et al 2013).

Rugby: In the domain of rugby, pose estimation and tracking are critical for performance analysis, injury prevention, and tactical assessments. One notable open-access dataset is the Kalisteo Rugby Sevens Dataset¹⁰, which provides annotated video sequences captured from actual Rugby Sevens matches (Maglo et al 2022). This dataset includes multiple match recordings with manually annotated player positions, offering a valuable resource for training and evaluating pose estimation and tracking models. Additionally, research efforts have been directed toward improving rugby scene classification and risk assessment using pose estimation techniques. Nonaka et al (2024) investigated the application of vision-language models to enhance rugby scene classification, utilizing a dataset of 18,000 labeled images extracted from 100 minutes of rugby match footage. This dataset includes key in-game scenarios such as scrums, mauls, rucks, tackles, and lineouts, which are crucial for developing robust pose estimation models. Furthermore, Nonaka et al (2022) proposed an end-to-end high-risk tackle detection system, leveraging pose estimation to assess tackle risks. Their system processes rugby match videos to identify frames containing tackles, localizes the tackle events, estimates player poses, and evaluates the associated risks.

Track and Field: In track and field sports, pose estimation and motion tracking are vital tools for performance evaluation, injury mitigation, and biomechanical studies. Researchers rely on annotated datasets to train and validate models for these tasks. The Leeds Sports Pose (LSP) Dataset¹¹, a foundational resource, contains 2,000 images of athletes in diverse sports, including track and field, each labeled with 14 joint positions for 2D pose analysis. Another key resource, the MPII Human Pose Dataset¹², expands this scope with 25,000 images and 40,000 annotated subjects, offering 2D and 3D labels to study dynamic ath-

⁹ <https://www.kaggle.com/competitions/nfl-big-data-bowl>

¹⁰ https://kalisteo.cea.fr/wp-content/uploads/2022/04/README_R7.html

¹¹ <https://paperswithcode.com/dataset/lsp>

¹² <https://paperswithcode.com/dataset/mpii-human-pose>

letic motions. For 3D movement analysis, the Human3.6M Dataset¹³ stands out, providing 3.6 million 3D poses across actions like running and jumping, crucial for modeling track and field kinematics. Specialized sports dynamics are captured in the SportsPose Dataset¹⁴ (Ingwersen et al 2023), which includes 176,000 3D poses from 24 athletes performing sprints and jumps, addressing the need for high-intensity motion data.

Swimming: In the field of swimming, pose estimation and tracking are crucial for performance analysis, technique optimization, and injury prevention (Giulietti et al 2023; Einfalt and Lienhart 2020). One of the most prominent datasets is SwimXYZ¹⁵, a large-scale synthetic dataset designed to address the challenges of underwater motion capture. It comprises approximately 3.4 million frames with 2D and 3D joint positions, along with 240 sequences of swimming motions using SMPL parameters. The dataset incorporates variations in camera angles, lighting conditions, and swimmer appearances, making it a valuable resource for pose estimation and motion tracking in aquatic environments (Fiche et al 2023). Another widely used dataset is the Annotated Dataset for Swimmer Pose Estimation¹⁶, which includes over 1,200 annotated video frames of swimmers performing the backstroke. Each frame is manually labeled with 14 key joint locations, providing detailed pose data essential for training and validating pose estimation models (Greif and Lienhart 2009). Additionally, the CADDY Underwater Stereo-Vision Dataset (Gomez Chavez et al 2019) contributes valuable underwater stereo-vision data, which, although not exclusively focused on swimming, can be used for human pose estimation in aquatic settings. This dataset includes recordings of divers performing various activities, with synchronized IMU data to provide accurate motion ground truth.

Rowing: Currently, no large-scale, standardized, open-access datasets for rowing motion analysis. One of the most frequently referenced datasets is the Rowing Motion Dataset (Wilaikaew et al 2023), which contains video sequences of rowers performing training exercises, annotated with pose estimation keypoints and stroke classification labels. Similarly, the Intelligent Pose Recognition and Evaluation System for Rowing Sports Dataset (Zhang et al 2024b) consists of motion capture data, synchronized video sequences, and labeled keypoints, making it valuable for pose estimation research and athlete movement tracking applications. This dataset has been utilized to evaluate rower performance and refine technique. Another notable dataset is the Inaccurate Action Detection Dataset for Rowing Machine Exercises (Chang et al 2024), which includes 665 correct and 490 erroneous rowing action sequences, each containing 200 angle data points. Designed for AI-based real-time motion accuracy detection, this dataset allows machine learning models to identify incorrect rowing techniques and provide automated feedback for training optimization.

Martial Arts: Pose estimation and tracking play a crucial role in martial arts by enabling robust analysis of body movements, stances, and transitions without requiring high-detail joint-level precision. The MADS (Martial Arts, Dancing, and Sports) Dataset (Zhang et al 2017) is a large-scale multimodal dataset designed for human action recognition, focusing on complex and dynamic movements in martial arts, dance, and sports. It contains high-quality video recordings captured from multiple viewpoints, along with corresponding

¹³ <https://paperswithcode.com/dataset/human3-6m>

¹⁴ <https://paperswithcode.com/dataset/sportspose>

¹⁵ <https://g-fiche.github.io/research-pages/swimxyz/>

¹⁶ https://www.uni-augsburg.de/de/fakultaet/fai/informatik/prof/mmc/research/research_projects/swimmer_pose_estimation_dataset/

motion capture (MoCap) data, pose annotations, and other metadata. The HardMo (Hard-case Motion) Dataset (Liao et al 2024) is a large-scale motion capture dataset specifically designed to address challenges in monocular human mesh recovery, particularly in complex dance and martial arts scenarios. Comprising over 7 million images with precise 2D key-points and 3D SMPL annotations, HardMo encompasses 15 dance categories and 14 martial arts categories, providing a diverse range of dynamic movements.

Cycling: Pose estimation and tracking are essential in cycling for analyzing rider posture, pedaling efficiency, and body stability. The Tsinghua-Daimler Urban Pose Dataset (TDUP)¹⁷ is a large-scale human pose estimation dataset focusing on vulnerable road users (VRUs) in urban traffic environments. Collected from high-resolution (at least 1920 x 1024 pixels) on-board cameras navigating complex Chinese urban streets, the dataset comprises approximately 21,000 images featuring around 90,000 manually labeled VRU instances.

Badminton: Pose estimation and tracking play a vital role in badminton by enabling efficient analysis of player movement, stance, and stroke execution. The Badminton Olympic Dataset (Ghosh et al 2018) comprises 10 videos sourced from YouTube, each showcasing individual badminton matches with an average duration of one hour. This dataset is enriched with diverse annotations, including bounding boxes marking player positions in 1,500 frames, notations of 751 instances where players score points, and comprehensive annotations delineating time boundaries and stroke labels across 12 stroke categories, such as serves and lobs. The VideoBadminton dataset (Li et al 2024c) is a specialized collection designed to advance action recognition in badminton. Curated from high-quality footage of 19 skilled players from National Central University's badminton team, the dataset encompasses 7,822 video clips totaling 145 minutes, covering 18 distinct stroke types such as smashes, drives, and net shots.

Table Tennis: Pose estimation and tracking of coarse-grained poses focus on overall body posture, arm-leg coordination, and footwork. The TTStroke-21 ME22 dataset (Martin et al 2023) is a specialized collection designed to advance action detection and classification in table tennis. It comprises 100 minutes of player-centered videos recorded at 120 frames per second, totaling over 718,000 frames. The dataset includes 20 distinct table tennis stroke classes, such as serves, offensive strokes, and defensive strokes, each meticulously annotated by professional players. The OpenTTGames dataset (Voeikov et al 2020) is an open resource designed to advance computer vision tasks in table tennis, including ball detection, semantic segmentation of players, tables, and scoreboards, as well as rapid in-game event detection. It features full-HD videos recorded at 120 frames per second, totaling 5 training videos (10 to 25 minutes each) and 7 shorter test videos. The P^2A (Ping Pong Action) dataset (Bian et al 2022) is a comprehensive resource designed to advance dense action detection in table tennis. It comprises 2,721 video clips, each approximately six minutes long, totaling 272 hours of footage sourced from professional table tennis matches, including the World Table Tennis Championships and the Olympics. The dataset features 139,075 labeled action segments across 14 distinct classes, such as serves, forehand and backhand strokes, smashes, and blocks. Each action segment is meticulously annotated with precise temporal boundaries, facilitating both action localization and recognition tasks.

Tennis: Pose estimation and tracking are essential in tennis for analyzing player movements, posture, and stroke execution. By focusing on overall body posture, limb coordination, and footwork, this approach allows for efficient monitoring of key actions such as

¹⁷ <https://fabian-flohr.de/datasets>

serves, smashes, and foot positioning, which are crucial for performance in these fast-paced sports. The Pose2Trajectory dataset (AlShami et al 2023) contains detailed information extracted from tennis videos, including 2D keypoints corresponding to the joints of the players and the ball's coordinates in each frame. These videos were sourced from the TennisTV channel on YouTube, featuring matches from the Vienna Open, a professional tennis tournament played on indoor hard courts. The Tennis Player Actions Dataset for Human Pose Estimation (2024a) is a specialized collection aimed at advancing human pose estimation in tennis. It comprises 2,000 images categorized into four distinct actions: backhand shot, forehand shot, ready position, and serve. Each image is annotated with 18 keypoints corresponding to human joints, following the COCO dataset format.

Archery: Pose estimation and tracking can be used to detect key body landmarks in archery. They help evaluate whether an archer maintains a consistent form across multiple shots, ensuring proper alignment of the body, bow, and target. Chiam et al (2023) introduced a foundational analytical model for quantitatively assessing an archer's performance. This model leverages a deep learning-based pose estimation technique to detect key human joints during elite archery shooting. The joint angle formed by the right shoulder, right elbow, and left wrist was then tracked and visualized relative to video frames. To build the dataset, 10 elite archers with a minimum of three years of training were recruited, with five using recurve bows and the other five using compound bows. Lee et al (2024) introduced an innovative approach for automatically analyzing archery shooting time using artificial intelligence and computer vision, with a focus on the crucial anchoring-to-release phase. The study utilized data from publicly available archery match broadcasts on YouTube, including events from the 2024 Paris Olympics and the Korean National Archery Team Special Match, enabling a detailed analysis of world-class competition environments and athlete movements.

5.2 Micro-action datasets

These sports involve intricate and highly detailed movements (fine-grained), often requiring precision at the level of specific body parts (e.g., fingers, wrists, or toes). These are commonly technique-focused sports where subtle variations can significantly impact performance. This subsection presents pose estimation and tracking datasets in fine-grained sports.

Gymnastics: Pose estimation and tracking can be applied at a fine-grained level to analyze precise body movements, joint angles, and postural stability during complex routines in gymnastics. The FineGym dataset (Shao et al 2020) is a comprehensive resource designed to advance fine-grained action recognition in gymnastics. It comprises 4,883 videos across various gymnastics events, offering a rich and diverse collection for research purposes. A distinctive feature of FineGym is its hierarchical annotation system, which categorizes actions into a three-level semantic hierarchy (events, sets, and elements). Researchers have utilized this dataset to develop and evaluate methods in action recognition, temporal action localization, and human-object interaction detection.

Figure Skating: Pose estimation and tracking can be used at a fine-grained level to analyze intricate body movements, joint angles, and precise alignment during routines in figure skating. The FSD-10 dataset (Liu et al 2020b) is a specialized resource designed for fine-grained action recognition in figure skating. It comprises 1,484 video clips sourced

from the 2017-2018 worldwide figure skating championships, encompassing 10 distinct actions performed by both male and female skaters. The dataset is meticulously annotated by experts, detailing aspects such as action type, execution grade, and skater information. The FineSkating dataset (Shan 2020) consists of 46 figure skating competition videos from 2018 and 2019, each approximately one hour in length. The dataset features a hierarchical labeling structure, including event and action labels. There are seven event categories, such as jumps and spins, with each event containing multiple specific actions. Additionally, each action is annotated with its start time, end time, and performance score, making the dataset suitable for both action recognition and action quality assessment. The Motion-Centered Figure Skating (MCFS) dataset (Liu et al 2021) is a specialized resource designed for fine-grained semantic analysis and temporal action segmentation in figure skating.

Diving: Pose estimation and tracking can be used at a fine-grained level to analyze body alignment, rotation angles, and entry precision throughout each phase of a dive in diving. The Diving48 dataset (Li et al 2018) is a fine-grained video collection designed for action recognition in competitive diving. It comprises approximately 18,000 trimmed video clips, each depicting one of 48 distinct diving sequences standardized by the International Swimming Federation (FINA). These sequences encompass various combinations of take-off techniques, somersaults, twists, and flight positions, capturing the nuanced complexities of diving performances.

Yoga: Pose estimation and tracking can be used at a fine-grained level to analyze precise joint angles, flexibility, and muscle engagement during asanas in yoga. The Yoga-82 dataset (Verma et al 2020) is a large-scale image dataset designed for yoga pose classification. It consists of 28,370 images (from different angles, lighting conditions, and settings) categorized into 82 different yoga poses, covering a broad range of asanas. The dataset includes images collected from diverse sources, including online platforms, ensuring a wide variety of poses performed by practitioners of different skill levels, backgrounds, and environments. The Yoga-107 dataset¹⁸ is a comprehensive collection designed for yoga pose classification tasks. It comprises images spanning 107 distinct yoga postures, offering a broad spectrum of asanas for analysis. The dataset is organized into separate folders for each pose, facilitating structured access and utilization. The 3D-Yoga dataset (Li et al 2022) is a comprehensive resource designed for visual-based hierarchical sports action analysis, particularly focusing on yoga poses. It comprises over 3,792 action samples and 16,668 RGB-D key frames, collected from 22 subjects performing 117 distinct yoga poses.

Ballet and Dance: Pose estimation and tracking can be used at a fine-grained level to analyze intricate body movements, posture precision, and fluidity of transitions in ballet and dance. The Dance-12 dataset¹⁹ is a curated collection designed for fine-grained dance posture recognition, encompassing 12 distinct dance styles. It comprises a total of 4,000 images, with each dance style represented by approximately 333 images. The dataset is organized to facilitate the study of complex body movements and poses inherent to various dance forms.

Some sports, like tennis or swimming, may involve both fine-grained and coarse-grained aspects depending on the focus of analysis. For example, in tennis, a serve (fine-grained) involves wrist movement, while running for a ball (coarse-grained) emphasizes overall

¹⁸ <https://www.kaggle.com/datasets/arowe/yoga-poses-dataset-107>

¹⁹ <https://sites.google.com/view/syd-net/home>

motion. In swimming, the stroke technique (fine-grained) can be analyzed alongside the general body motion through water (coarse-grained). The classification might also depend on the role or action within the sport, such as a goalkeeper in soccer (fine-grained movements like catching) versus an outfield player (coarse-grained running and kicking). Table 5 summarizes the sports according to their action granularity.

5.3 Ground-truth techniques

Creating a high-quality ground-truth dataset is a time-consuming and iterative process that needs careful planning and attention to detail. The most common techniques for creating ground truth data for pose estimation in sports are described in this subsection.

Motion Capture Systems (MoCap) are one of the most accurate methods for capturing detailed ground-truth pose data. These systems use specialized hardware (e.g., cameras, markers, or sensors) to track the 3D positions of an athlete's joints in real-time (Zhang et al 2024a). MoCap has two techniques: marker-based and markerless. In former systems, reflective markers are placed on the athlete's body at specific anatomical landmarks (e.g., joints), and cameras track the 3D position of these markers (Romero-Flores et al 2024). Later systems use cameras and sophisticated algorithms to estimate the position of joints without requiring physical markers on the body (Jatesiktat et al 2024).

Video-based Annotation (2D and 3D) tools can be used to create ground-truth annotations. They have less resource-intensive setups. These methods involve manually annotating frames in video sequences to create labeled datasets, which can then be used to train and evaluate models. According to the dimension, there are two kinds of annotation tools. In former tools, manual or semi-automatic placement of keypoints (e.g., head, shoulders, elbows, etc.) is annotated in 2D images. In later systems, depth information along with 2D positions is annotated to generate 3D poses (Shen et al 2023).

Wearable sensors, particularly IMUs, can be used to create ground-truth data by directly measuring the orientation and position of different body parts during movement. These sensors capture motion through accelerometers, gyroscopes, and magnetometers (Moniruzzaman et al 2023).

Synthetic data can be generated using computer simulations or 3D rendering software in situations where collecting real-world ground-truth data is challenging or expensive (Black et al 2023). 3D rendering software simulates human movement and annotates joints for training models. Physics-based simulations simulate realistic human movement using physical models to create diverse datasets with varied poses (Greff et al 2022).

Table 5 Classification of sports into fine-grained and coarse-grained categories

Category	Characteristics	Examples
Fine-grained sports	Focus on intricate, highly detailed movements involving small body parts such as fingers, wrists, or toes.	Archery, Gymnastics, Badminton, Table Tennis, Tennis, Golf, Diving, Figure Skating, Fencing, Ballet, Dance
Coarse-grained sports	Emphasis on larger, full-body movements, focusing on general motion patterns rather than fine details.	Soccer, Basketball, Football (American), Rugby, Track and Field, Swimming, Rowing, Martial Arts, Cycling
Overlapping cases	Certain sports involve both fine-grained and coarse-grained actions depending on the context or specific actions.	Tennis (serve vs. running), Swimming (stroke technique vs. body motion), Soccer (goalkeeper's catching vs. running).

Crowd-sourcing can be used to accelerate the creation of large annotated datasets by distributing the task of labeling data to a large group of people. This can be particularly helpful for 2D keypoint annotation. Human annotators manually label keypoints or poses in videos, either through direct annotation or through gamified tasks. Also, crowd-sourcing with semi-automatic systems suggests initial annotations which are later refined by human annotators (Cormier et al 2021). Table 6 summarizes the ground-truth data creation techniques.

6 Case studies and real-world applications in sports

The integration of deep learning and computer vision has had a transformative impact on the field of HPE and pose tracking, making it a crucial tool in the domain of sports science. HPE systems facilitate real-time analysis of the three-dimensional movements of athletes, allowing precise measurement of biomechanical parameters such as joint angles, movement velocities, and force distributions. This level of detail allows for the identification of performance constraints, including asymmetries, muscular weaknesses, and deficiencies in coordination. By leveraging these insights, practitioners can design personalized training regimens, correct technical flaws, and implement targeted injury prevention strategies. Furthermore, HPE and pose tracking offer an objective framework for performance evaluation, empowering coaches and rehabilitation specialists to optimize training outcomes and accelerate recovery processes. This section gives the case studies and real-world applications related to pose estimation and tracking in sports.

6.1 Action recognition, prediction, detection, and tracking

Action recognition aims to accurately classify a specific movement or activity performed by an athlete, while action prediction involves predicting future movements based on current position and motion dynamics. For instance, a football player's passing motion, a tennis player's serving position, or a gymnast's complex figures are all examples of sport-specific movements that can be recognized. In particular, deep learning-based methods such as CNN and Spatio-Temporal Graph Convolutional Networks (ST-GCN) are highly effective in identifying these movements from video data (Rahevar et al 2023; Pang et al 2023). Action prediction applications are used in strategic decision-making processes in fast-paced sports, such as predicting the direction of a pass in a football match or predicting the shooting point of a basketball player. Time-based models such as Temporal Convolutional Networks (TCN) and LSTM networks are widely used in such predictions (Sun et al 2024b; Xi et al 2024).

Action detection and tracking are also widely used and important in real-world sport applications. Action detection is used to determine when and where certain movements occur in a sequence of events. Detection systems supported by pose estimation can automatically determine the start of a serve in a tennis match, the accuracy of a gymnastic movement, or the moment of a goal in a football match. The system is employed particularly in the context of referee decision-support systems and player performance analysis (Wu and Fan 2024; Zhao 2024). The development of action tracking was driven by the need to monitor the movements of athletes and objects over time continuously. This has led to the creation of applications that can be used to measure the running distance of players in foot-

Table 6 Ground-truth data creation techniques for pose estimation and tracking in sports

Technique	Description	Tools/Systems	Example
MoCap systems	Tracks body movements in 3D using markers or cameras.	Vicon ^a , OptiTrack ^b , Qualisys ^c	Tracking basketball player movements (e.g., jumps, ball handling).
Video-based annotation	Manual or semi-automatic labeling of 2D/3D pose landmarks from video frames.	LabellImg ^d , CVAT ^e , DeepLabCut ^f	Annotating soccer players' joint positions during a game.
Wearable sensors	Captures motion data through sensors placed on body parts.	Xsens ^g , Shimmer ^h , Myotest ⁱ	Monitoring swimming strokes using wearable sensors to capture joint angles and posture.
Synthetic data generation	Uses computer simulations or 3D rendering software to generate labeled pose data.	Blender, Unity3D ^j , Unreal Engine ^k	Creating synthetic tennis player movements for pose estimation model training.
Crowdsourcing (Semi-automatic)	Distributes annotation tasks to a large number of people or uses semi-automatic labeling.	Amazon Mechanical Turk ^l , Supervise.ly ^m	Crowdsourcing annotation of track and field runners' poses during various events.

^a<https://www.vicon.com/>

^b<https://www.optitrack.com/>

^c<https://www.qualisys.com/>

^d<https://github.com/HumanSignal/labellimg>,

^e<https://www.cvat.ai/>

^f<https://github.com/DeepLabCut/DeepLabCut>

^g<https://www.movella.com/products/xsens>

^h<https://shimmersensing.com/>

ⁱ<https://www.myotest.com/>

^j<https://unity.com/>

^k<https://www.unrealengine.com/>

^l<https://www.mturk.com/>,

^m<https://supervise.ly.com/>

ball, examine defensive and offensive strategies in basketball matches, or track the speed and path of the ball in tennis matches. The most commonly used algorithms for tracking player and object movements include the Kalman filter, the particle filter, and DeepSORT (Wu and Fan 2024; Akan and Varlı 2023; Liu et al 2024a). Table 7 provides action recognition, prediction, detection, and tracking in sports applications. It shows the practical impact and effectiveness of recent approaches in HPE and pose tracking applications across sports. Guo et al.'s (2021) 95.2% accuracy in action recognition and Pang et al.'s (2023) 96.4% accuracy clearly demonstrate the strong capability of deep learning-based approaches like CNN and ST-GCN models. Similarly, the 89.06% accuracy of Xu et al.'s (2024) predictive methods demonstrates considerable potential for informing strategic decisions in dynamic sports environments. Detection methods achieve accuracies around 88.7%, providing critical support for referee decision-making and precise event identification (Wu and Zhou 2024). Tracking methods emphasize continuous real-time analysis capabilities essential for performance monitoring and tactical evaluation, though these are less explicitly quantified. These numerical results reinforce the growing importance and utility of advanced HPE and pose tracking methods, confirming their vital role in enhancing athlete training, competitive analysis, and sport-specific decision support systems.

6.2 Sport performance analysis and tracking

HPE and tracking technologies are employed as cutting-edge instruments to enhance individual performance and examine team dynamics in sports. In comparison to conventional techniques, HPE and tracking enable the monitoring of athletes' movements in their natural environment through the utilization of markerless technology, thereby facilitating a more comprehensive and accurate performance assessment than conventional techniques (Guo et al 2021). HPE provides the opportunity to examine and quantify movement data in detail in the biomechanical analysis of athletes. In particular, measuring kinematic parameters such as joint angles, speed, and acceleration provides various information to improve the technical efficiency of athletes. For example, stride analysis of runners can reveal optimization opportunities to increase running economy, while biomechanical analysis of racket speed in tennis athletes helps address deficiencies in stroke techniques (Xu et al 2024). It also continuously monitors critical parameters of the athlete, such as posture, movement speed, and muscle activations, allowing technical errors to be corrected in real-time (Rahevar et al 2023). To illustrate, in the context of football matches, the running positions of players can be subjected to immediate analysis, thereby facilitating the implementation of defensive adjustments. Similarly, in the domain of gymnastics, instances of balance loss or faulty movements can be addressed through the provision of prompt corrective guidance during training sessions, thus facilitating the improvement of performance (Toshpulatov et al 2024). Furthermore, it serves as a crucial instrument for the observation of equipment and ball movements throughout the game, irrespective of the players involved. It is used to conduct strategic analyses in ball sports such as tennis, volleyball, basketball, and football. In basketball, the speed and trajectory of the ball can be observed and evaluated to assess the efficacy of offensive strategies. Similarly, in tennis, the path and velocity of the ball after a serve can be analyzed to ascertain the effectiveness of the subsequent shots (Alsawadi 2024). Table 8 presents the expanding scope and practical impact of HPE and tracking methods in sports performance analysis.

Table 7 Overview of action recognition, prediction, detection, and tracking in sports

Category	Subcategories	Description	Methods	Advantages	Challenges	Performance
Recognition (Guo et al 2021; Rahevar et al 2023; Sun et al 2022; Pang et al 2023)	Action Type Classification, Sport-Specific Movements	Classifying actions such as shot, pass, or serve.	CNN, ST-GCN, Attention Networks	Accurate classification, Personalized analysis	Overlapping movements, Data limitations	Generally 85-98% accuracy (e.g., Guo: 95.2%, Pang: 96.4%, Sun: 84.7 (NTU-RG-BD)%, Rahevar: 88.2%)
Prediction (Xu et al 2024; Liu et al 2020a)	Future Pose Prediction, Player/Object Motion Forecasting	Predicting future actions based on pose dynamics.	LSTM, TCN, Transformers	Strategy development, Real-time simulations	Latency issues, Complex dynamics	Typically 85-90% accuracy (e.g., Xu: 89.06% (NTU-60), Liu: high accuracy)
Detection (Ding et al 2020; Yang et al 2023; Wu and Zhou 2024)	Critical Movement Detection, Timing/Position Identification	Identifying when and where actions occur.	YOLO, Faster R-CNN, Attention Models	Detects key movements, Real-time analysis	Occlusions, Low-quality frames	Mostly 80-90% accuracy (Wu&Zhou: 88.7% accuracy, 84.4% F1-score (Taekwondo), similar in other studies)
Tracking (Toshpulatov et al 2024; Ma et al 2021)	Movement Monitoring, Multi-Target Tracking	Tracking movements of athletes and objects over time.	DeepSORT, GNN, Kalman Filters	Long-term analysis, Player tracking	Speed-accuracy tradeoff, Multi-target complexity	It is usually highly accurate and real-time. (Specific numerical data is not specified.)

Sport performance tracking involves the systematic collection and analysis of data related to an athlete's physical and technical performance, with the goal of enhancing training, strategy, and overall outcomes. Using tools such as wearable sensors, GPS devices, motion capture systems, and video analysis, coaches and athletes can monitor metrics like speed, heart rate, movement efficiency, and fatigue levels in real-time or post-session. This data-driven approach not only aids in optimizing individual and team performance but also plays a crucial role in injury prevention, workload management, and personalized training regimens. As technology advances, sport performance tracking is becoming increasingly precise and accessible, revolutionizing the way athletes prepare and compete.

As detailed in the table, recent studies have reported substantial performance gains, with recognition accuracies reaching 95.2% (Guo et al 2021), motion prediction models achieving 89.06% (Xu et al 2024), and real-time feedback systems operating at speeds exceeding 100 FPS (Toshpulatov et al 2024). While explicit metrics are less frequently reported in equipment and ball tracking, the consistent use of DeepSORT and Kalman-based frameworks demonstrates tracking accuracies above 90% in controlled environments. These empirical results not only validate the reliability and effectiveness of HPE-based systems but also highlight their potential to enhance athlete monitoring, optimize performance, and support data-driven decision-making. Future studies should aim to address current limitations such as variability in real-world conditions and the lack of standardized benchmarks, thereby expanding the applicability of these technologies across diverse sporting contexts.

6.3 Broadcasting and visualization

The use of visualization technologies in sports broadcasting is becoming increasingly prevalent, offering viewers a more profound comprehension of the events they observe and a more engaging experience. There are various visualization applications such as heat maps, trajectory visualization, player position, and tracking systems. Heat maps are employed to illustrate the movement of players on the field and areas of high density. These maps graphically represent the distribution of player positions, thereby facilitating strategic analysis for both the technical team and the audience. In the context of football matches, for instance, heat maps can be employed to monitor the activity of players in defensive or offensive areas (Xu et al 2024). Trajectory visualization is an important tool that provides a graphical representation of the paths followed by athletes or equipment such as balls, making it easier for both technical teams and spectators to understand the dynamics of movement. HPE-based trajectory visualization uses deep learning methods such as CNN and GNN to analyze the motion paths in detail to understand the dynamic nature of sports. For example, the speed and direction of the ball in a basketball match are modeled with images from multiple camera systems, providing a strategic analysis of both the players' positions and the ball's motion (Guo et al 2021; Xu et al 2024). The future of trajectory visualization technologies will become more impressive with the integration of augmented reality (AR) and virtual reality (VR) platforms. Moreover, the integration of visual, sensor, and biomechanical data through multimodal data integration will facilitate more comprehensive analyses (Toshpulatov et al 2024). Furthermore, player position and tracking systems are employed for strategic analysis and performance evaluation in the context of team sports. The objective of these systems is to monitor and analyze the instantaneous movements of athletes on the field, including their speed, position, and their situation about other players.

Table 8 Real-world applications of HPE and tracking in sports performance analysis

Application	Methods	Advantages	Challenges	Applications	Performance
Biomechanics and motion analysis (Guo et al 2021; Xu et al 2024; Yeadon and Pain 2023)	ST-GCN, CNN, Markerless Motion Capture	Precise joint angle, velocity, and acceleration measurements; reduced injury risks; movement efficiency.	Occlusion in multi-athlete scenarios; high costs; calibration issues.	Running stride analysis; tennis swing optimization; gymnastics posture refinement; weightlifting technique assessment; yoga pose correction.	High accuracy across tasks (e.g., Guo: 95.2% recognition; Xu: 89.06% prediction). Validated models show strong biomechanical consistency (Yeadon-Pain).
Real-time feedback (Sun et al 2022; Toshpulatov et al 2024; Millet et al 2021)	Wearable Sensors, Visual Tracking, Augmented Reality	Instant feedback; enhanced decision-making; personalized performance improvement.	Real-time processing demands; environmental variability.	Player fatigue tracking in football; balance correction in gymnastics; sprint training adjustments for runners; real-time swing analysis in tennis; live feedback for basketball dribbling.	Typically high responsiveness (e.g., Toshpulatov: 95.2% accuracy, 106.9 V/s runtime speed; Sun: 84.7% on NTU-RGBD). Reliable real-time feedback validated (Millet).
Equipment and ball tracking (Alsawadi 2024; Verheul et al 2020)	Multi-Camera Systems, DeepSORT, Kalman Filters	Accurate motion analysis; game strategy enhancement; ball and player synchronization.	Resolution constraints; complexity in trajectory modeling.	Ball trajectory tracking in basketball; post-serve ball dynamics in tennis; cricket bowling speed analysis; baseball pitching mechanics evaluation; goal-line technology enhancement in football.	Generally accurate tracking with DeepSORT/Kalman filters (commonly 90% in controlled settings).

Table 9 presents technologies, benefits, challenges, and applications for broadcasting and visualization in HPE and tracking. As shown in Table 9, player activity mapping through heatmaps achieves recognition accuracies up to 95.2% (Guo et al 2021), while re-projection errors typically range between 2.7 px and 6.4 px, depending on both the employed model and the complexity of the scene. (Baumgartner and Klatt 2023). Trajectory visualization systems have shown clear improvements, with a 4.3% increase in accuracy (Xu et al 2024) and sub-centimeter 3D trajectory error (Chao et al 2023), supporting precise movement diagnostics. Furthermore, real-time augmented statistics reached operational capacities of 255 FPS, with 72.2 mAP and 91.5 AP50 on benchmark datasets (Chen et al 2022), enabling seamless integration into live broadcasts. Enhanced replay systems and dynamic tracking models reported < 9 cm RMSE in spatial localization (Blauberger et al 2021), ensuring reliability in fast-paced multi-player scenarios. These advancements clearly demonstrate the immense potential of HPE-based visualization tools in enriching audience engagement and delivering high-fidelity, data-driven insights for coaching, tactical evaluation, and automated event interpretation.

6.4 Injury prevention

Injuries are an inherent risk of participation in any sport, either at the individual or team level. Such injuries can have significant physical, psychological, and economic consequences (Bahr et al 2018; Helme et al 2021). A comprehensive understanding of the risk factors associated with injury and the way these factors interact is essential to prevent potential injuries during participation in sports activities (Van Eetvelde et al 2021; Leckey et al 2024). It is very important to predict the risk of sports injuries to prevent injuries, optimize athletic performance, and develop recovery strategies, as these factors are integral to the realm of sports activities. The existing literature demonstrates that accurately assessing injury risk enables the creation of individualized training and rehabilitation strategies, which facilitate both the minimization of the likelihood of injury and a more efficient recovery process when injuries occur (Vincent et al 2022; Musat et al 2024). Many studies are currently being conducted by HPE researchers to develop strategies for the prevention and early detection of injuries in a variety of sports. The sport of basketball often requires athletes to perform high-impact movements, including jumping, abrupt changes of direction, and physical contact. These actions can potentially cause a variety of injuries among athletes. HPE systems analyze joint flexion during landings to reduce anterior cruciate ligament (ACL) and ankle injuries, resulting in significant improvements in athlete safety and performance (Lu and Chang 2012). In football, risk factors for ACL injuries are identified through the assessment of knee alignment using a range of algorithms. It has been demonstrated that the implementation of neuromuscular exercises has been shown to lead to a marked reduction in the incidence of ACL injuries (Bahr and Engebretsen 2011). The monitoring and analysis of balance during high-intensity routines with HPE has been shown to be an effective method to reduce wrist and elbow injuries, which are prevalent in the field of gymnastics (Hernández-Beltrán et al 2023; Farana et al 2023). In the field of athletics, it is used to prevent the onset of iliotibial band syndrome (IT) and shin discomfort. This is achieved through a comprehensive assessment of step symmetry and the influence of fatigue on the musculoskeletal system, which then guides the development of targeted corrective training protocols (Ray et al 2019). Table 10 shows that HPE-based injury prevention techniques have advanced sports medi-

Table 9 Enhanced broadcasting and visualization in HPE and tracking

Feature	Technologies	Advantages	Challenges	Applications	Performance
Player heatmaps (Guo et al 2021; Baumgartner et al 2023; Dai and Baumgartner 2023)	HPE AI-gorithms, Multi-Camera Systems	Visualizes player activity zones; Analyzes team dynamics and strategies; Identifies high-intensity areas.	Requires accurate tracking; High computational demand; Data integration complexities.	Football: Movement heatmaps; Basketball: Defensive zone coverage; Volleyball: Serve positioning analysis; Cricket: Fielding zones.	Guo et al. (2021) achieved 95.2% accuracy in karate action recognition; Baumgartner et al., (2023) re-projection errors in broadcast 3D pose estimation typically range from 2.7 px to 6.4 px.; Dai (2023) optimized camera setup improved 3D realism and coverage.
Trajectory visualizations (Xu et al 2024; Suo et al 2024; Toshpulatov et al 2024; Ding et al 2020)	Kalman Filters, Spatiotemporal Graph Neural Networks, DeepSORT	Displays ball and player paths; Enhances play dynamics understanding; Provides real-time trajectory analysis.	Data quality dependency; Real-time latency issues; Complexity in crowded scenarios.	Tennis: Ball trajectory post-serve; Football: Shot tracking; Cricket: Bowling dynamics; Baseball: Outfield throws.	Xu et al. (2024) reported a 4.3% accuracy gain; Suo et al. (2024) showed high visual fidelity without metrics; Toshpulatov et al. (2024) found DDC3N superior to GCNs; Ding et al. (2020) achieved SOTA with STG-Ins.
Real-time augmented stats (Chen et al 2022; Mahmood et al 2017; Lo et al 2022; Soltani and Morice 2020)	Augmented Reality (AR), Wearable Sensors	Provides real-time player statistics; Enhances audience engagement; Integrates analytics into broadcasts.	Latency in visual rendering; Requires seamless hardware-software integration; Environmental limitations.	Football: Real-time sprint speeds; Tennis: Serve speed analytics; Basketball: Pass accuracy overlay; Swimming: Stroke frequency tracking.	Chen et al. (2022) reported 72.2 mAP and 255 FPS; Mahmood (2017) enabled low-light face recognition at 5x5 px; Lo (2022) and Soltani (2020) showed AR improves understanding and skill correction.
Enhanced replay systems (Baumgartner and Klatt 2023; Chao et al 2023)	3D Reconstruction, AI-Based Video Processing	Provides detailed motion analysis; Highlights critical moments; Enables interactive viewing options.	Requires advanced computational hardware; Time-consuming rendering for high-detail frames; Dependency on accurate data.	Football: Goal-line replays; Tennis: Serve legality; Baseball: Strike zone analysis; Rugby: Tackle efficiency visualization.	Baumgartner (2023) reported 2.44 cm 3D error and 2.76 px reprojection error; Chao (2023) achieved 0.23 cm 3D trajectory error with calibrated camera reconstructions.
Dynamic player position tracking (Kovalchik 2023; Blauberger et al 2021; Zhang et al 2020)	GPS, IMUs, HPE with Multi-Camera Systems	Tracks player movements accurately; Enhances tactical evaluations; Monitors real-time positioning.	High resource requirements; Environmental variability; Challenges in occluded scenarios.	Football: Team formation visualization; Volleyball: Defense alignment; Basketball: Fast-break tracking; Hockey: Line rotations.	Blauberger (2021) reported <9 cm RMSE; Zhang (2020) reduced ID switches using the DeepPlayer model; Kovalchik (2023) extracted spatiotemporal metrics and roles without reporting explicit errors.

cine by enabling precise monitoring and biomechanical analysis. These methods have been shown to support injury prevention efforts, particularly in ACL injuries and ankle sprains, by enhancing movement analysis and risk factor identification through biomechanical and AI-assisted approaches (Plakias et al 2024; Yang et al 2024). Real-time biomechanical feedback and predictive modeling based on joint-angle analyses (Abrams et al 2016) further enhance the capability to preemptively address injury risks. Specialized AI-driven models for gymnastics and gait (Panconi et al 2024; Sowmia et al 2023) have enabled personalized injury prevention strategies, optimizing athlete safety and performance. These technologies make significant contributions to preventive sports healthcare and highlight opportunities for future research in personalized, sport-specific injury prevention frameworks.

6.5 Refereeing and game analytics

Those involved in the field of sports, including coaches, managers, media personnel, pundits, and fans, have historically sought to provide the most accurate and comprehensive explanations of the events that occur on the field of play. Until recently, broadcast footage served as the primary source of information, leading to the development of observational qualitative analysis (Vidal-Codina et al 2022). In particular, recent years have witnessed a significant impact of developments in computer vision and AI on the field of sports analytics, with notable advances in the areas of refereeing and game analysis. Technologies such as pose estimation, player tracking, and ball tracking are becoming indispensable tools for enhancing the precision and efficacy of refereeing, while also facilitating a more profound comprehension of game strategies (Cao et al 2017). Nowadays, automatic systems for foul detection, player positioning, and ball tracking are being researched to assist referees and optimize their decision-making processes (Siddiqui et al 2024; Shao and He 2025). These innovations have the potential to reduce human error, accelerate decision-making processes, and provide real-time data for tactical analysis, training, and performance evaluation (Liu et al 2023b). Table 11 provides information for foul detection, ball tracking, player positioning, and event detection. As shown in Table 11, pose-based refereeing and analytics systems have demonstrated measurable improvements in accuracy and reliability across sports applications. Cao et al (2017) achieved leading results on COCO and MPII datasets for foul detection, while Li (2022) and Du et al (2024) improved classification via vision-based learning, despite lacking exact metrics. In ball tracking, Vicente-Martínez et al (2023) reported 95% accuracy with YOLOv7 + DeepSORT, and Blauburger et al (2021) achieved 2D RMSE of 15 cm and 3D errors of 17–21 cm under real match conditions. Liu et al (2024a) reached 98.23% pose accuracy at 16.7 FPS for volleyball player tracking. Event detection models, such as those by Xiao et al (2023) and Hashmi et al (2023), reported up to 98.91% accuracy, 97.8% precision, and 98.1% F1-score. Collectively, these advances illustrate the efficacy of HPE in enhancing automated decision-making, improving fairness, and enabling detailed game analysis.

Table 10 HPE-based injury prevention techniques

Sport	Technique	Injury types	Applications	Benefits
Football (Soccer) (Plakias et al 2024)	Gait and load monitoring	ACL injuries, stress fractures	Tracks knee valgus during cutting movements and monitors running load to reduce biomechanical stress.	Provides a comprehensive bibliometric analysis of football biomechanics, highlighting key research trends in injury prevention and performance analysis.
Basketball, Handball, Volleyball (Milić et al 2025)	injury epidemiology and prevention strategies	Ankle sprains, knee ligament injuries (especially ACL), shoulder overuse injuries	Injury prevention planning, gender- and position-specific risk analysis, athlete monitoring	Highlights injury patterns and supports the use of proprioceptive training for reducing ankle and knee injuries.
Tennis (Abrams et al 2016)	Shoulder rotation analysis	Rotator cuff injuries, tennis elbow	Tracks serving biomechanics to detect and minimize overuse injuries.	Prevents chronic shoulder injuries; improves swing efficiency.
Gait (Walking) (Panconi et al 2024; Yang et al 2024)	Markerless pose estimation using DeepLabCut	Gait-related disorders; general lower-limb injury monitoring	Clinical gait assessment, rehabilitation, remote monitoring	Improves temporal gait analysis accuracy; reduces need for lab equipment.
Gymnastics (Sowmia et al 2023; Vu-Minh et al 2024)	Joint load and dynamic stability monitoring	Wrist and elbow injuries, ACL injuries	Monitors repetitive joint stress and balance stability during routines.	Helps maintain correct posture during intense workouts; reduces injury risk; acts as a virtual trainer.
Running (Johnston et al 2019; Salian et al 2024)	Fatigue monitoring and stride symmetry evaluation	Shin splints, IT band syndrome	Evaluates stride mechanics and fatigue-related biomechanical changes to reduce overuse injuries.	Enhances running efficiency; minimizes injury risks.

7 Challenges and future research directions

Although HPE and tracking technologies show great potential in sports, they are faced with several substantial challenges. These challenges include difficulties in adapting to different sports and changing conditions (Nath et al 2019; Mathis and Mathis 2020; Gong et al 2016; Zhang et al 2023a; Badiola-Bengoa and Mendez-Zorrilla 2021). Moreover, there is a paucity of annotated data for a range of sports (Stenum et al 2021b), concerns regarding the collection of data in accordance with privacy regulations, and ethical considerations pertaining to the limitations of the datasets (Cioppa et al 2024). Consequently, to achieve comprehensive integration of HPE and tracking systems in sports, further research is necessary to enhance the model's adaptability to varying scenarios, augment real-time processing speed (Gong et al 2016; Munea et al 2020), surmount occlusions, and accurately analyze intricate movements. This section explains challenges and then gives future research directions in pose estimation and tracking in sports.

7.1 Challenges

Pose estimation and tracking in sports are constrained by numerous technical and methodological challenges that significantly impact the scalability and reliability of the process. Technical challenges include managing occlusion during complex interactions (Xi et al 2024), correcting inaccuracies arising from low-resolution or noisy input data (Ferrara 2024), and achieving real-time analysis despite limited computational resources (Rehman et al 2024). These challenges are particularly pronounced in high-speed sports, where achieving both precision and low latency is critical.

Additionally, dataset limitations pose a substantial barrier. Current datasets often lack diversity in terms of sports disciplines, athlete demographics, and environmental conditions (Ren et al 2025), leading to reduced robustness and limited generalizability of trained models (Knap 2024). This issue is compounded by generalization challenges, as models developed for specific scenarios frequently fail to adapt effectively to new or unseen conditions. Integrating multimodal data presents another layer of complexity, as combining visual, sensor, and biomechanical data necessitates advanced methods to synchronize and process heterogeneous information (Hu et al 2024). This subsection presents these challenges, examining their implications for the field and identifying opportunities for innovative advancements to overcome these limitations.

7.1.1 Technical challenges

Occlusion and overlapping poses are critical challenges in pose estimation and tracking in sports. Occlusion occurs when parts of a player's body are obscured by other players, equipment, or environmental objects, resulting in incomplete or ambiguous input data. In team sports such as basketball, football, or rugby, where physical interactions and dense player formations are common, severe partial or full occlusions can lead to significant degradation in the accuracy of pose predictions (Citraro et al 2020). This issue is further compounded by overlapping poses, where multiple players occupy similar spatial locations in the field of view. Traditional pose estimation models, particularly those relying on single-frame analysis, struggle to disentangle individual poses in such scenarios due to the lack of spatial

Table 11 Pose based refereeing and game analytics in sports

Topic	Description	Technologies	Challenges	Applications	Performance
Automatic foul detection (Cao et al 2017; Li 2022; Du et al 2024)	Automatic foul detection aims to identify fouls based on player movements, body interactions, and contact in real-time using advanced computer vision and AI models. These systems help reduce human error and assist referees in making accurate decisions.	OpenPose, AlphaPose, Machine Learning, Pose Estimation Models, Deep Learning	Accuracy in detecting subtle fouls, real-time processing in fast-paced sports, occlusion, distinguishing different types of fouls	Used in sports like Basketball, Football, and Rugby for automatic foul detection, aiding referees during matches to ensure fair play and consistent rulings	Cao (2017) achieved top benchmark scores (COCO, MPII); Li (2022) and Du (2024) reported improved foul detection.
Ball tracking (Vicente-Martinez et al 2023; Blauberger et al 2021)	Ball tracking involves monitoring the ball's position, speed, and trajectory in real-time. It uses object detection techniques such as YOLO and Faster R-CNN to enhance decision-making, strategy analysis, and performance evaluation.	YOLO, Faster R-CNN, Object Detection, Computer Vision Algorithms, Deep Learning	Occlusion, tracking accuracy during fast movements, sensitivity to environmental conditions (lighting, camera angles), real-time processing	Widely applied in Tennis (Hawk-Eye), Football, Basketball, and other sports to track the ball's trajectory and position, aiding in strategic analysis and decision-making	Vicente-Martinez (2023) reported 95% tracking accuracy using YOLOv7 combined with DeepSORT; Blauberger (2021) observed 2D RMSE of 15 cm and 3D errors ranging from 17 to 21 cm in dynamic scenes.
Player positioning (Blauberger et al 2021; Liu et al 2024a; Klatt et al 2021)	Player positioning refers to the tracking of player movements on the field using pose estimation technologies. This allows for tactical analysis and performance evaluation by understanding player behavior and positioning.	MediaPipe, OpenPose, Pose Estimation Models, Player Tracking, AI Integration	Tracking multiple players simultaneously, occlusion, real-time analysis of multiple interactions, distinguishing player positions in crowded situations	Used in Football, Basketball, and Soccer for tactical analysis, fitness tracking, performance evaluation, and strategic planning	Liu (2024) achieved 98.23% pose estimation accuracy at 16.7 FPS in volleyball tracking; Klatt (2021) focused on the visual behavior and positioning of basketball referees.
Event detection (Tsagkatakis et al 2017; Xiao et al 2023; Hashmi et al 2023)	Event detection systems aim to automatically identify significant events during a game, such as goals, fouls, or assists, by analyzing video footage and real-time data.	Computer Vision, Event Detection Algorithms, Real-Time Video Analysis	Handling multiple types of events, detecting subtle game moments, ensuring real-time accuracy	Event detection in sports like football, basketball, and tennis is increasingly automated to provide insights into key moments during games, assisting in refereeing and post-game analysis	Xiao (2023) reported up to 98.9% accuracy in activity recognition; Hashmi (2023) achieved 97.8% precision and 98.1% F1-score in event detection; Tsagkatakis (2017) demonstrated robust goal detection performance.

separation or unique visual cues. Additionally, occlusions that persist over multiple frames introduce temporal inconsistencies, further complicating tracking tasks (Zhang et al 2017; Arif et al 2022).

To address these challenges, recent research has focused on the development of spatiotemporal models that leverage motion continuity across video frames to infer missing pose components during occlusion. Transformer-based architectures and GCNs have shown promise in representing relationships between body parts and individuals within a shared scene. Multi-camera systems and 3D reconstruction techniques also provide avenues for mitigating occlusion by combining data from diverse viewpoints (Zhang et al 2017; Nibali et al 2021). However, these approaches often come with trade-offs in computational complexity and real-time feasibility, highlighting the need for lightweight, efficient solutions tailored to sports applications.

Advances in occlusion-robust algorithms, such as those employing adversarial training or synthetic occlusion during data augmentation, could further enhance model performance. Table 12 provides a consolidated evaluation of key technical challenges in sports pose estimation, with a strong focus on occlusion, overlapping subjects, low resolution, and temporal instability, conditions frequently encountered in dynamic game environments. Notably, Zecha et al (2019) achieved up to 97.1% PCK by incorporating ILP-based rectification, highlighting the effectiveness of optimization-driven correction in occluded frames. Arif et al (2022) demonstrated 90% precision using ROI-based attention models in cluttered scenes, while Zhang et al (2017) quantified errors between 100-200mm, emphasizing the need for improved matching strategies. The table also underscores the role of novel sensor integration: Hasegawa et al (2022) improved landmark visibility under low light via IR-assisted methods. Temporal consistency was enhanced by Fastovets et al (2013) through smoothing techniques, and Bridgeman et al (2019) validated multi-frame tracking under real-time match conditions. The table's primary contribution lies in bridging empirical performance data with targeted technical responses. It not only benchmarks current capabilities but also guides future system design by highlighting which solutions yield measurable gains. As an actionable resource, it encourages further integration of multi-modal inputs, adaptive filtering, and attention-based learning to mitigate remaining limitations in occluded or complex multi-player contexts.

7.1.2 Dataset limitations

Dataset limitations for pose estimation and tracking in sports present significant challenges that hinder the development of accurate and robust models. A key issue is the lack of sports-specific annotated data, as most existing datasets are general-purpose and fail to represent unique movements such as those in gymnastics, swimming, or martial arts (Seong et al 2024; Kaseris et al 2024). This limits the ability of models to generalize effectively across diverse sports. Additionally, privacy and ethical concerns arise when collecting sensitive athlete data, especially in real-world training or competitive settings. These concerns necessitate the adoption of privacy-preserving technologies such as federated learning and anonymization techniques (Liu et al 2024b).

Another limitation is the imbalance in sports representation, where popular sports like football and basketball dominate datasets, leaving niche activities underrepresented (Kaseris et al 2024). This skew restricts model versatility and calls for balanced dataset development

Table 12 Technical challenges with occlusion and overlapping players

Challenge	Description	Impact	Solutions	Performance
Occlusion (Zecha et al 2019; Arif et al 2022; Bai et al 2024)	Players or objects block parts of the body, creating missing or ambiguous data.	Reduced pose accuracy, inconsistent tracking, and degraded performance in dense environments.	Spatiotemporal models, occlusion-aware training, and multi-camera setups.	Zecha (2019) reported up to 97.1% PCK using a rectification pipeline; Arif (2022) achieved 90% precision with ROI-attention models; Bai (2024) improved occluded keypoint detection using HRNet combined with COHead.
Overlapping players (Zhang et al 2017; Bridgeman et al 2019; Nibali et al 2021)	Close proximity between players causes difficulty in distinguishing individual poses.	Misassociation of keypoints, inability to track individual players accurately.	Graph-based pose models, association algorithms, and pose affinity fields.	Zhang (2017) reported errors ranging from 100–200 mm in challenging sequences; Bridgeman (2019) and Nibali (2021) improved keypoint identification in overlapping scenarios.
Low-resolution data (Hasegawa et al 2022; Slembrouck et al 2023)	Limited resolution from cameras results in unclear body part keypoints.	Poor estimation of small or intricate movements, especially in fast sports like basketball.	Super-resolution preprocessing, multi-scale feature extraction, and high-definition cameras.	Slembrouck et al. (2023) demonstrated that engagement and compliance were maintained despite the limited mobility; Hasegawa (2022) showed that an IR-assisted model enhanced landmark detection in low-light conditions.
Real-time processing (Fastovets et al 2013; Bridgeman et al 2019; Bai et al 2024)	Fast-paced games require low latency, but complex models are computationally expensive.	Trade-off between accuracy and speed impacts live feedback applications like referee support.	Lightweight deep learning models, hardware acceleration (e.g., GPUs), and real-time optimization frameworks.	Fastovets et al (2013) enhanced temporal stability in pose sequences through keyframe-guided models and spatiotemporal smoothness constraints; Bridgeman (2019) validated real-time feedback in match settings; Bai (2024) showed that optimized models maintained accuracy with low latency.
Lighting variability (Jafarzadeh et al 2021; Nibali et al 2021; Slembrouck et al 2023)	Changes in lighting (e.g., shadows, low light, bright sunlight) introduce noise into visual data.	Reduced pose detection accuracy, particularly in outdoor sports or poorly lit indoor arenas.	Lighting-augmented datasets, adaptive normalization, and IR or thermal cameras for low-light conditions.	Jafarzadeh et al. (2021) improved landmark stability in low light; Nibali et al. (2021) enhanced robustness with attention; Slembrouck et al. (2023) maintained accuracy using SR across lighting.
Dataset Diversity (Zecha et al 2019; Nibali et al 2021; Bai et al 2024)	Datasets lack representation for different sports, demographics, and environments.	Models trained on specific sports or scenarios fail to generalize across new contexts.	Creation of diverse datasets (e.g., ASPset), synthetic data generation, and transfer learning techniques.	Zecha et al. (2019) reached >96% PCK with about 8500 frames; Nibali et al. (2021) improved generalization via diversity; Bai et al. (2024) boosted occlusion robustness with OCPose.
Perspective issues (Zhang et al 2017; Citraro et al 2020; Baumgartner and Klatt 2023; Dai et al 2024; Souza et al 2024)	Variations in camera angles and viewpoints disrupt the consistent tracking of poses.	Errors in 3D pose reconstruction and multi-camera alignment.	Multi-view geometry methods, perspective transformation networks, and 3D pose refinement techniques.	Zhang et al. (2017) reduced error to 100–200 mm via bi-directional matching; Citraro et al. (2020) and Baumgartner (2023) lowered reprojection error to about 2 cm with multi-view fusion; Dai et al. (2024) and Souza et al. (2024) improved 3D realism using optimized camera layouts.

Table 12 (continued)

Challenge	Description	Impact	Solutions	Performance
Dynamic backgrounds (Fastovets et al 2013; Arif et al 2022; Bai et al 2024)	Movements of other players, spectators, or objects in the scene add complexity to pose estimation tasks.	False positives in key-point detection and instability in pose tracking.	Background subtraction, region-of-interest (ROI) extraction, and attention-based deep learning models.	Arif et al. (2022) achieved 90% precision with ROI-attention; Fastovets et al (2013) improved temporal stability; Bai et al. (2024) enhanced tracking with CONet.
Temporal consistency (Fastovets et al 2013; Bridgeman et al 2019; Zecha et al 2019)	Pose estimation across consecutive frames may result in jittery or inconsistent predictions in videos.	Poor motion tracking and disrupted downstream applications like biomechanical analysis.	Temporal smoothing, graph convolutional networks, and recurrent neural networks (RNNs) for temporal integration.	Fastovets et al (2013) improved temporal stability using spatio-temporal filtering; Zecha et al. (2019) enhanced continuity with ILP-based correction; Bridgeman et al. (2019) accelerated multi-frame tracking with fast geometric matching.

Table 13 Dataset limitations for pose estimation and tracking in sports

Limitation	Description	Impact	Solutions	Performance
Lack of sports-specific annotated data (Cioppa et al 2024; Suo et al 2024)	Existing datasets are general-purpose, lacking specificity for unique sports actions such as gymnastics, swimming, or martial arts.	Limits model generalization to diverse sports; reduces accuracy in analyzing domain-specific movements.	Develop domain-specific datasets; employ synthetic data generation and transfer learning to supplement dataset creation.	Cioppa et al. (2024) introduced SoccerNet-Depth with 12.4k frames; Suo et al. (2025) reported limited generalization due to annotation gaps; existing models showed reduced accuracy without sport-specific labels.
Privacy and ethical concerns (Chen et al 2020b; Hoppe et al 2018; Liu et al 2024b)	Capturing sensitive athlete data during training or competition raises privacy issues and legal challenges.	Restricts data collection opportunities; risks violating ethical standards and trust among athletes.	Implement federated learning and anonymization techniques; follow strict ethical guidelines for data collection.	Liu et al. (2024b) maintained low MAE/RMSE under sparse data with privacy-aware models; Chen et al. (2020) enabled secure prediction via privacy-preserving fusion; Hoppe et al. (2018) noted ethical risks without reporting metrics.
Imbalance in sports representation (Van Zandycke and De Vleeschouwer 2022; Colyer et al 2018; Pagnon et al 2021, 2022)	Popular sports like football dominate datasets, while niche sports (e.g., fencing, archery) lack representation.	Models underperform in niche sports due to insufficient training data; reduces versatility.	Create balanced datasets with diverse sports; use data augmentation and few-shot learning for rare sports scenarios.	Pagnon et al. (2021) found 50% accuracy loss with fewer cameras; Zandycke et al. (2022) noted only about 10% of data showed realistic basketball motion; Colyer et al. (2018) reached 2°–3° accuracy, limited to sagittal movements.
Lack of multimodal data (Nogueira et al 2025; Askari and Alavi 2023; Edwards 2016)	Current datasets focus on visual data, excluding sensor or biomechanical data such as IMU or EMG recordings.	Reduces depth of analysis; fails to capture the full biomechanical context of movements.	Combine visual data with multimodal sources; develop hybrid models capable of integrating heterogeneous data.	Nogueira et al. (2025) reported up to 23% accuracy loss without depth; Askari & Alavi (2023) achieved 76.6% with pose-only SSL; Edwards (2018) found single-modal inputs inadequate for fatigue detection.
Temporal Data Limitations (Bertasiu et al 2019; Pandya et al 2023; Gui et al 2018)	Static images dominate datasets, while temporal data needed for motion analysis is scarce.	Prevents accurate motion dynamics analysis; hinders understanding of temporal patterns in sports.	Develop video datasets annotated with temporal markers; train RNNs and Transformers for sequence analysis.	Bertasiu et al. (2019) achieved 88.7% mAP with temporal warping vs. 83.8% with optical flow; Pandya et al. (2023) noted 2-frame event drift from desynchronization; Gui et al. (2018) reported realism loss from error accumulation.
Scalability and cost constraints (Souza et al 2024; Zhang et al 2023a)	Large-scale dataset creation is resource-intensive, requiring significant time and cost.	Limits dataset size and diversity; increases reliance on small-scale, less representative datasets.	Use automated annotation tools; crowdsourcing can reduce costs and accelerate labeling.	Souza et al. (2024) noted that heat exceeding 45°C forced costly scheduling shifts, while Zhang et al. (2023a) reported that 3D pose models require over 80 labeled samples, limiting scalability.

Table 13 (continued)

Limitation	Description	Impact	Solutions	Performance
Data quality and consistency (Stenum et al 2021b; Elhayek et al 2016; Liu and Yuan 2018)	Variations in camera setups, lighting, and annotation practices result in noisy datasets.	Reduces model reliability; leads to inaccuracies and inconsistencies in training and validation phases.	Standardize data collection protocols; establish robust data cleaning pipelines.	Stenum et al. (2021b) reported up to 7.4° error in 2D pose vs. 3D; Elhayek et al. (2016) noted degraded multi-person tracking in poor-quality videos; Liu & Yuan (2018) found unclear heatmaps impair joint detection.
Cultural and demographic bias (Hoppe et al 2018; Dai and Baumgartner 2023)	Data is often biased towards athletes from specific regions or demographics, limiting generalization to diverse populations.	Fails to account for global diversity; reduces inclusivity and fairness in sports applications.	Expand datasets to include diverse cultural and demographic representations; train culturally sensitive models.	Hoppe et al. (2018) reported up to 20% accuracy loss on non-Western datasets; Dai & Baumgartner (2023) observed a 7.8° pose error with shifts in body shape and viewpoint.

and the application of data augmentation techniques. Moreover, the absence of multimodal data, such as biomechanical or sensor-based measurements, reduces the comprehensiveness of pose analysis (Al Mudawi et al 2024). Integrating multimodal data sources can enhance accuracy and broaden applications.

Temporal data limitations, arising from a focus on static images over video sequences, impede motion dynamics analysis (Mishra et al 2024). Developing annotated video datasets and employing temporal models like RNNs and Transformers can address these gaps. Additionally, the high cost and time required for large-scale dataset creation pose scalability issues, which can be mitigated through automated annotation tools and crowdsourcing. Variability in data quality and consistency due to differences in lighting, camera setups, and annotation practices further challenges model performance (Smerdov et al 2020). Standardizing collection protocols and establishing robust cleaning pipelines can resolve these inconsistencies.

Lastly, cultural and demographic biases limit the generalizability of models to diverse athlete populations, emphasizing the need for datasets that reflect global diversity (Seong et al 2024). Addressing these limitations through innovative solutions will ensure the development of more inclusive, accurate, and efficient sports pose estimation systems, advancing both research and real-world applications. Table 13 provides pivotal dataset gaps in sports pose estimation, revealing tangible performance drops, for instance, 23% lower accuracy with unimodal data (Nogueira et al 2025) and a 7.8° error under body-shape variance (Dai and Baumgartner 2023). Such shortcomings underscore the pressing need for domain-specific labeling (e.g., gymnastics or fencing), standardized multimodal captures (e.g., combining RGB, IMU), and culturally diverse sampling to address up to 20% lower accuracy on non-Western athletes (Hoppe et al 2018). In addition, resource-efficient annotation strategies (e.g., synthetic data, federated learning) could mitigate high development costs (Souza et al 2024). Together, this table highlights the necessity to design inclusive, specialized, and robust datasets to push the boundaries of both scientific insight and practical deployment in sports analytics.

7.1.3 Generalization issues

The HPE systems have shown remarkable advancements, particularly in applications such as activity recognition and biomechanical analysis. However, their performance can fluctuate significantly across different sports disciplines. This limitation is primarily attributed to the fact that many HPE models are trained on general-purpose datasets like COCO or MPII, which lack the specific nuances and diversity inherent to various sports. Each sport presents unique biomechanical patterns, movement dynamics, and environmental conditions, making it challenging for models trained on generic datasets to generalize effectively to sport-specific scenarios. For instance, models trained on a dataset with a high proportion of football data cannot be generalized to sports with different biomechanical patterns and environmental factors, such as basketball, volleyball, gymnastics, or swimming (Nath et al 2019).

When models are over-optimized for training data, they struggle to adapt to unknown data such as new camera angles, lighting conditions, or athlete physiologies, resulting in over-learning. Furthermore, motion capture systems trained in controlled laboratory environments perform poorly in dynamic outdoor sports (Chen et al 2020b; Mathis and Mathis

2020). In these cases, a wider variety of datasets must be designed to increase the generalizability of the models.

The majority of existing datasets comprise static images and motion data that lack continuity, which presents a significant challenge for the modeling of high-speed or complex movements, particularly those observed in gymnastics and basketball (Stenum et al 2021a; Badiola-Bengoia and Mendez-Zorrilla 2021). To overcome these problems, the use of time-stamped video datasets, RNNs, or transformer-based models is required. Table 14 provides a comprehensive and technically valuable overview of generalization challenges in sports pose estimation, systematically identifying seven key problem domains. These include limited training diversity, overfitting, environmental variability, biomechanical variation, and the absence of temporal or domain-adaptive strategies. The table not only documents the quantifiable impacts, such as accuracy drops >5 px, joint angle errors up to 7.4° , or 7.3% improvement via LSTM integration, but also proposes targeted solutions. Crucially, the table highlights how context-specific biases, such as domain gaps or underrepresented populations, directly translate to performance degradation in real-world sports settings. Its structured alignment of challenges, impacts, and solutions makes it a valuable resource for guiding the design of more robust, adaptable, and inclusive HPE systems. In general, this synthesis contributes significantly to closing the gap between lab-based model development and real-world athletic applications.

7.2 Future research directions

The field of pose estimation and tracking in sports has advanced significantly with the advent of deep learning and computational techniques. However, there are still several open challenges and potential research directions that can further enhance performance, applicability, and robustness. We structure future research in pose estimation and tracking in sports into the following actionable dimensions.

7.2.1 Multi-modal approaches

Integrating multi-modal data sources for pose estimation and tracking in sports offers significant potential to improve accuracy, robustness, and generalization. By leveraging diverse information, such as RGB video, depth maps, and sensor data, multi-modal approaches can overcome the limitations of single-modal methods. Future directions on multi-modality can be listed as follows. Robust techniques can be developed to combine RGB video, depth maps, thermal imaging, and IMUs effectively (Guo et al 2024). Pose estimation performance can be enhanced in scenarios with poor lighting, occlusion, or high-speed motion using complementary information from multiple modalities (Yan et al 2023). Adaptive systems can prioritize or switch modalities based on environmental and contextual requirements. Combining multimodal pose estimation with biomechanical models is critical for detailed motion analysis and sports-specific insights (Guidolin et al 2024). Some concrete research questions on modalities and data integration are as follows:

- How can modality-specific noise (e.g., motion blur in RGB, drift in IMUs) be mitigated during fusion?
- Can sensor selection be automated based on environmental factors (e.g., lighting, oc-

Table 14 Generalization issues in pose estimation and tracking in sports

Challenge	Description	Impact	Solutions	Performance
Lack of diverse training data (Nath et al 2019)	Datasets are often narrowly focused on specific sports or controlled conditions. For example, datasets dominated by football may fail to generalize to other sports like swimming or gymnastics.	Limits generalization to new sports, environments, or unseen data.	Develop diverse and inclusive datasets representing multiple sports and demographics.	Achieved human-level accuracy (2.7 px error) with 500 frames; accuracy significantly drops (>5 px error) when applied to untrained species or behaviors due to limited training diversity.
Overfitting (Chen et al 2020b; Mathis and Mathis 2020)	Models become overly specialized for training data, reducing adaptability to real-world scenarios. Overfitting occurs when models are too optimized for the training data, leading to reduced performance in dynamic settings.	Reduces performance in new contexts, such as varying camera angles or lighting conditions.	Regularize models; employ data augmentation to simulate diverse scenarios.	Chen et al. (2020b): Models trained on limited data exhibit up to 15% lower accuracy on unseen datasets. Mathis & Mathis (2020): Transfer learning and data augmentation with as few as 100 frames reduced pose estimation errors to less than 5 pixels, effectively mitigating overfitting.
Environmental variability (Gong et al 2016; Zhang et al 2023a; Zhao 2023)	Training data often lacks diversity in lighting, backgrounds, or weather conditions. For instance, most datasets do not capture the variability in outdoor sports or challenging lighting	Models underperform in dynamic environments like outdoor sports or under low-light conditions.	Use synthetic or augmented data to replicate varied environmental conditions.	Gong et al. (2016): Accuracy drops under lighting and background changes. Zhang et al. (2023a): CNN-LSTM improves stability by 15%. Zhao (2023): Hypergraph model boosts accuracy by 10% in variable environments.
Biomechanical variations (Stenum et al 2021b; Badiola-Bengoa and Mendez-Zorrilla 2021)	Differences in athlete physiques and movement patterns are underrepresented in datasets. For example, gender-specific or demographic-specific variations in motion are often overlooked.	Results in biased or inaccurate pose estimation for certain demographics or sports.	Collect data from diverse populations; apply transfer learning for adaptation.	Stenum et al. (2021b): Hip, knee, and ankle angles showed errors of 4.0°-7.4° vs. motion capture. Badiola-Bengoa & Mendez-Zorrilla (2021): General-purpose HPE systems underperform in complex sports with non-standard body movements.
Temporal data limitations (Tejero-de Pablos et al 2018; Muneca et al 2020)	Static images dominate datasets, lacking sequences to capture motion dynamics. Without temporal data, models struggle to track high-speed actions like running or ball handling.	Reduces the accuracy of tracking complex sports movements.	Develop video datasets with temporal annotations; use RNNs and Transformer models.	Tejero-de-Pablos et al. (2018): Combining 2D/3D action features with LSTM improved highlight detection accuracy by 7.3% over CNN-only models. Muneca et al. (2020): Lack of temporal modeling in 2D HPE leads to inconsistent keypoint tracking, reducing accuracy in multi-frame analysis.

Table 14 (continued)

Challenge	Description	Impact	Solutions	Performance
Domain gap (Mathis and Mathis 2020; Zhang et al 2023a)	Training data from controlled environments differs from real-world sports settings. For example, lab-based motion capture data does not replicate real-world sports conditions.	Hinders model performance when applied to real-world sports applications.	Use domain adaptation techniques to minimize discrepancies between training and target environments.	Mathis & Mathis (2020): Pretraining on ImageNet improves generalization; fewer than 100 labeled frames yield <5 px error via transfer learning. Zhang et al. (2023a): Semi-supervised CNN-LSTM model trained on 80 samples shows strong generalization in dynamic sports scenes.
Data quality and noise (Gong et al 2016; Stenum et al 2021b)	Training data often includes inaccuracies or inconsistencies in annotations. These errors, such as mislabeling or missing keypoints, undermine the reliability of predictions.	Leads to errors in predictions, reducing reliability and robustness of models.	Standardize annotation protocols; use robust data cleaning methods.	Gong et al. (2016): Illumination, occlusion, and background noise degrade estimation stability. Stenum et al. (2021b): OpenPose showed sagittal joint errors of 4.0°-7.4° vs. motion capture in 2D gait analysis.

clusion)?

- How can biomechanical priors be integrated into deep learning models for explainable insights?

7.2.2 Transformer models

Transformer models have emerged as a powerful paradigm in computer vision, including pose estimation and tracking, due to their ability to handle complex spatial and temporal relationships (Lin et al 2022). Future directions on transformer models can be listed as follows. Lightweight and efficient transformer models can be tailored for pose estimation and tracking in real-time sports scenarios (Liang et al 2024; Chen et al 2022). Transformers capturing local and global (multiscale representation) features can be developed for accurate motion analysis in dynamic sports environments. Transformers may be generalized across various sports disciplines with minimal fine-tuning (Li et al 2024b; Chen et al 2024). For richer predictions, transformer-based models can be extended to handle and fuse data from multiple modalities, such as RGB video, depth maps, and IMUs. Some concrete research questions on model architectures and learning paradigms are as follows:

- What design trade-offs exist between accuracy and latency in sports-specific transformer models?
- How can pre-trained transformer backbones be adapted with minimal fine-tuning across sports domains?
- Can self-supervised or contrastive learning improve generalization to unseen sports?

7.2.3 Edge computing and real-time systems

Edge computing and real-time systems are transformative technologies for estimating and tracking poses, especially in sports. They enable low-latency, efficient, and scalable solutions for applications that demand immediate insights and decision-making (Bozzini et al 2024). Future directions on edge computing and real-time systems can be listed as follows. It is important to design computationally efficient models for real-time pose tracking on resource-constrained edge devices and wearables for personalized sports performance monitoring and injury prevention (Zhu et al 2024; Singh et al 2024; Zhou et al 2024). Innovative methods may reduce processing delays, enabling ultra-low-latency applications for live sports analytics and broadcasting (Francisco et al 2024). Hybrid edge-cloud architectures combine edge and cloud computing for complex tasks, processing critical real-time data locally while offloading less urgent tasks to the cloud (Saha et al 2024). Also, edge systems can be standardized for seamless integration across different sports technologies and platforms. Some concrete research questions on deployment and real-time systems are as follows:

- How can we compress models without compromising on pose fidelity for wearable deployment?
- What are optimal cloud-edge partitioning strategies for different sports use cases?
- Can real-time feedback systems be personalized for injury prevention or tactical adjustment?

7.2.4 LLMs' role in enhancing pose estimation and tracking

Large Language Models (LLMs), such as GPT, are primarily designed for text-based tasks but can significantly contribute to enhancing pose estimation and tracking in sports by complementing computer vision and machine learning models (Feng et al 2024). Future directions on LLMs can be listed as follows. LLMs can encode domain-specific knowledge, such as rules of a particular sport, to interpret pose estimation outputs in a meaningful way. LLMs can assist in generating text-based annotations for pose datasets, explaining movements or actions, which reduces the burden of manual labeling (Li et al 2024d). LLMs, combined with vision models, can facilitate multi-modal learning by integrating pose data with descriptive text, such as commentary or instructions, for enhanced understanding of movements. By processing natural language queries, LLMs can tailor pose analysis to specific requirements, such as analyzing a specific athlete or play (Wang et al 2024b). Some concrete research questions on multimodal interaction with LLMs are as follows:

- How can LLMs be used to interpret pose sequences into semantically meaningful descriptions?
- Can LLMs act as weak supervision agents by generating pseudo-labels or feedback?
- How do we evaluate and benchmark vision-language integration specifically for sports analysis?

7.2.5 Ethical and privacy considerations

Pose estimation and tracking in sports offer transformative capabilities for performance analysis, injury prevention, and fan engagement. However, these advancements raise significant ethical and privacy concerns that must be addressed to ensure responsible use (Andrews et al 2024). Future directions on ethical and privacy considerations can be listed as follows. Techniques like federated learning, anonymization, and differential privacy approaches can be considered to ensure data privacy and regulatory compliance while retaining its utility for analysis (Ahmad et al 2024; Fu et al 2024). Clear protocols must be established for obtaining informed consent from athletes, ensuring they understand how their data will be used and processed. Addressing algorithmic bias by using diverse and inclusive datasets is important for training pose estimation models. Robust encryption and secure storage infrastructures prevent data breaches and unauthorized access (Liu et al 2024b). Some concrete research questions on ethical and privacy-first research design are as follows:

- How do federated learning approaches perform across diverse sports environments?
- What metrics and datasets are needed to assess fairness across gender, body types, and ethnicities?
- How can consent and transparency mechanisms be embedded in sports tech platforms?

7.2.6 Action granularity

Action granularity plays a pivotal role in the effectiveness of pose estimation and tracking systems in sports. As technology continues to evolve, higher levels of action granularity will help create more sophisticated systems, further enhancing the support provided to ath-

letes, coaches, analysts, and fans (Gu et al 2024). Future directions on action granularity can be listed as follows. It is important to design adaptive systems that can switch between different levels of granularity depending on the context of the action being analyzed (e.g., focusing on specific body parts or motions during key moments in a game). Different levels of action granularity (e.g., from macro to micro-movements) can be combined to provide a more comprehensive understanding of athlete performance (Li et al 2024a; An et al 2024). Improved visualization techniques enhance visualization tools that represent granular motion data effectively, allowing coaches, athletes, and analysts to easily interpret and act on the information provided. Some concrete research questions on action granularity and interpretability are as follows:

- How can micro-actions (e.g., subtle hand shifts in table tennis) be detected in real-time?
- What role can temporal attention mechanisms play in distinguishing similar actions (e.g., jab vs. cross)?
- Can hierarchical models offer top-down (play level) and bottom-up (joint-level) interpretations?

8 Conclusion

Pose estimation and tracking in sports has rapidly evolved into a critical area of research, enabling transformative applications in performance analysis, injury prevention, and tactical decision-making. This survey contributes a structured and comprehensive synthesis of existing methodologies, datasets, and challenges, while identifying key trends and research gaps in the domain. By systematically comparing traditional techniques with modern deep learning-based approaches, particularly in complex, multi-person, and dynamic environments, this work provides a consolidated reference for both newcomers and experienced researchers.

A key practical takeaway from this survey is the identification of sports-specific constraints—such as occlusions, rapid motion, and inter-player interactions—that require specialized model designs and data augmentation strategies. Our analysis reveals the pressing need for curated, diverse, and high-resolution datasets that reflect real-world sports conditions. To enable real-time applications, researchers should prioritize lightweight, energy-efficient models optimized for edge deployment. Furthermore, the ethical implications of large-scale tracking, including privacy and consent, must be proactively addressed by integrating fairness-aware design and transparent data governance practices.

Looking ahead, promising directions include leveraging multimodal sensor fusion (e.g., combining video with IMU or biomechanical data), applying transformer-based architectures to capture long-range dependencies, and developing cross-sport generalization frameworks. We also encourage the creation of benchmark competitions and standardized evaluation protocols tailored to sports contexts, which would drive reproducibility and collaboration within the community. Ultimately, advancing this field will require interdisciplinary partnerships—bridging computer vision, sports science, and ethics—to design robust, adaptive, and responsible pose tracking systems.

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